
A Computationally Feasible Framework for Causal Probabilistic Explanation

Rafal Urbaniak*
Basis Research Institute

Eli Bingham†
Basis Research Institute

Poorva Garg‡
University of California, Los Angeles

Jack Feser§
Basis Research Institute

Daniel Waxman¶
Basis Research Institute

Abstract

Causal attribution and explanation are mainstays of philosophical, scientific, and policy analysis. Some current frameworks, however, such as actual causality, suffer from an inability to scale causal reasoning to large problems, while attribution methods that do scale—such as SHAP or LIME—are not causally aware. To address this two-fold problem, we present Probabilistic Causal Impact (PCI). Inspired by Halpern and Pearl’s actual causality and Pearl’s probability of necessary and sufficient causation, PCI provides tractable causal explanations in causal probabilistic machine learning models. By the same token, PCI provides explanations for events in the world, insofar as the model adequately represents real mechanisms. By re-conceptualizing explanation as an estimation problem on an expanded, causal probabilistic model, PCI allows for the use of well-established approximation algorithms, unlocking computationally tractable, truly causal explanations for large models. We test PCI on scaled-up, but canonical test cases for actual causation involving overdetermination and undercutting, finding that we recover results deemed intuitive by the literature, and illustrate that the correct behavior is preserved in the continuous version of the problem. We apply PCI to a real-life causal spatio-temporal model with complex machine-learned components trained on millions of data points. In this setting, we compare PCI to the non-causal (yet scalable) attribution method SHAP, and find that our approach yields explanations that agree with domain expert opinion, while SHAP does not.

Keywords: Probabilistic Causal Models, Explanation, Responsibility, Actual Causality, Probability of Necessity and Sufficiency.

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*rafal@basis.ai
†eli@basis.ai
‡poorvagarg@cs.ucla.edu
§jack@basis.ai
¶dan@basis.ai

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1 Introduction

Causal attribution is a central reasoning task in many important problems, such as gauging the efficacy of medical treatments [Shepherd et al., 2024], pinpointing drivers of economic growth [Cheshire and Magrini, 2009], root cause analysis in automated systems [Papageorgiou et al., 2022], or assessing the impact of public policy [Heckman and Pinto, 2022]. We may want to understand why an adverse outcome occurred so as to prevent recurrence, or how a predicted positive outcome could be brought about.

In the absence of scalable, formal causal attribution tools, analysts often resort to ad-hoc methods — strong correlation, historical comparison, domain expertise, or non-causal attribution like Granger causality [Granger, 1969] and SHAP [Lundberg and Lee, 2017]. Even researchers with an advanced command of statistical and experimental methods may not possess a thorough understanding of how to ascribe cause [Shepherd et al., 2024], and would benefit from scalable automation. Here we present a tool for accurate and tractable causal attribution applicable to a wide array of causal, probabilistic, machine-learned models.

Current methods for attributing causes to specific outcomes, such as actual causality [Halpern, 2016], are intractable in general, with no clear approximation [Eiter and Lukasiewicz, 2002]. Explanation methods that *do* scale are either non-causal [Lundberg and Lee, 2017] or amount to local, gradient-based sensitivity analyses [Butler et al., 2022]. SHAP [Lundberg and Lee, 2017] grounds attribution in observational correlations, so a feature can receive high attribution merely for being correlated with a true cause. Causal SHAP [Heskes et al., 2020] addresses this to some extent, but some failure modes persist (Section 4). Butler et al. [2022]’s Differential Causal Effect (DCE) relies on gradients, which requires model differentiability and does not capture effects from non-infinitesimal changes in potential causes. (DCE is, fairly, not designed for responsibility attribution: Butler et al. target rates of change of treatment.)

We introduce *probabilistic causal impacts* (PCI), an attribution framework that combines the context-sensitivity of actual causality with the scalability of probability-of-causation methods. PCI generalises Pearl’s probability of necessary and sufficient causation [Pearl, 2022] from binary to arbitrary variable domains, and incorporates the *witness* mechanism from actual causality [Halpern, 2016] via integration over a distribution of witness sets rather than existential quantification.

PCI also makes partial contact with the literature on *descriptive normality* [Halpern and Hitchcock, 2015, Icard et al., 2017]: attributions are computed as expectations against a distribution over counterfactual worlds conditioned on upstream factual context, a concrete operationalisation of the normality intuition. Several components have precedents — expectations over exogenous noise [Halpern, 2016], weighting by alternative-value probabilities [Beckers and Vennekens, 2015], aggregation over witness sets [Beckers and Vennekens, 2015, Halpern, 2016] — but no prior construction combines them into a single tractable estimator handling arbitrary variable domains.

PCI presumes access to a trained probabilistic causal model from which counterfactual outcomes can be sampled — a stronger requirement than methods that work from observational data and a causal graph alone (e.g., PN/PS/PNS bounds, Causal SHAP). Within that requirement, however, we place no strong restrictions on the form of the model, distributions, or types of variables. Such models can comprise standard machine learning components such as neural networks or Gaussian processes.

With this breadth of application, and a measure that surfaces its design choices — the variable selection distribution, alternative value distribution, and causal impact function — as explicit user-facing components rather than hidden defaults, PCI hopes to provide causal attribution and explanation tooling for a wide array of users and applications.

We validate PCI along four lines:

1. **Theoretical connection to actual causality.** Under suitable choices of Γ and Δ , PCI recovers Halpern’s actual-causality verdicts exactly (Section ??). A more extensive conceptual discussion appears in Appendix ??.
2. **Recovery on canonical actual-causality failure cases.** On scaled-up overdetermination and undercutting examples, PCI produces attributions matching the literature’s intuitive judgements (Section ??); the same holds in the continuous version (Section ??).

3. **Scalability to ML-grade models.** PCI produces individual-level attributions on a causal spatio-temporal model with machine-learned components trained on millions of data points (Section ??), where SHAP yields qualitatively different explanations.
4. **Comparison to gradient-based attribution.** Against Differential Causal Effect [Butler et al., 2022], PCI better reveals non-infinitesimal causal effects (Appendix ??).

2 Causal Impact: motivations

The central difficulty in causal attribution is not computing counterfactuals — it is knowing which counterfactuals to compute, and recognising when a cause is blocked by context rather than absent altogether. The closest predecessors of this approach, taken individually, fail to capture this: actual causality handles context sensitivity but is computationally intractable and not general enough; the probability of necessity and sufficiency scales well, but lacks context sensitivity. This section builds intuition for both limitations through a running example, setting up the design choices that motivate PCI in Section 3.

2.1 Actual Causality

Consider the following (deterministic, discrete) model:

Example 1 (Bob at Old Boys’ Club Bank). *Bob applied for a home mortgage loan with the local branch of Old Boys’ Club Bank (OBCB), and his application was denied. The OBCB policy requires applicants to pass a credit score check, but it only performs this check if the applicant is male; all non-male applicants are rejected outright. Thus, the procedure is:*

$$check = (gender = male)$$

$$check-failed = check \wedge (credit = bad)$$

$$loan = \neg check-failed \wedge (gender = male)$$

Why was Bob’s loan application denied? An intuitive explanation is that his credit was bad—had it been good, the outcome would have been different. In contrast, his gender does not appear to be responsible for the rejection: had it been different, his application still would have been denied (albeit this time due to $check = F$).

This suggests a counterfactual approach to feature responsibility attribution: perhaps we can identify the features responsible for a particular outcome by asking counterfactual questions—would the outcome have been different if a specific feature had a different value? This is called the *but-for clause*: Bob wouldn’t have been denied the loan but for his bad credit.

Concretely, consider a model M with *exogenous variables* $\mathbf{U}$, which are the source of stochasticity, and *endogenous variables* $\mathbf{V}$, which are influenced by other variables in the model. Let $\mathbf{S} \subseteq \mathbf{V}$ be the variables we suspect to be causally responsible (call them *causal suspects*), and $Y \in \mathbf{V}$ be an outcome of interest. Suppose we consider only one suspect $S \in \mathbf{S}$ and mark factual values with $*$. Under a setting $\mathbf{U} = \mathbf{u}$ of the exogenous variables, M produces $S = s^*$ and $Y = y^*$, denoted $(M, \mathbf{u}) \models S = s^*, Y = y^*$. We call this the *factual scenario*. An intervened model where a variable S is set to an alternative value s' is denoted $(M, \mathbf{u})[S \leftarrow s']$.

A first attempt at defining a counterfactual notion of causal responsibility is:

Definition 2 (But-for, single). *$S = s^*$ is causally responsible for $Y = y^*$ in $(M, \mathbf{u})$ if and only if $(M, \mathbf{u}) \models S = s^*, Y = y^*$ and $(M, \mathbf{u})[S \leftarrow s'] \not\models Y = y^*$ for some $s' \neq s^*$.*

In a given context, S is causally responsible for an outcome only when intervening on S with some alternative value would change that outcome. Bob’s credit score satisfies this condition: had his credit score been higher, the loan might have been approved. However, this cause test is too coarse.

Consider another case:

Example 3 (Alice at OBCB). *Alice is a woman with bad credit. She applies for a loan with OBCB and is rejected.*

Why was Alice denied the loan? Under the but-for clause for single cause candidates, neither her gender nor her credit score are responsible. No matter how high her credit score, her application

would still have been denied because the bank does not loan to women. Had her gender been different, she would still have been denied for having bad credit.

To address this example, we could consider sets of factors as causes:

Attempt 1 (But-for, sets). *Variables $\mathbf{C} \subseteq \mathbf{S}$ with factual values $\mathbf{C} = \mathbf{c}^*$ are causally responsible for $Y = y^*$ in $(M, \mathbf{u})$ if and only if $(M, \mathbf{u}) \models \mathbf{C} = \mathbf{c}^*, Y = y^*$ and $(M, \mathbf{u})[\mathbf{C} \leftarrow \mathbf{c}'] \not\models Y = y^*$ for some $\mathbf{c}'$ everywhere different from $\mathbf{c}^*$, where no proper subset of $\mathbf{C}$ has this property.*

The minimality condition reflects our preference for explanations that include only relevant properties.

Under this generalization, the set $\mathbf{c}^* = \{\text{gender} = \text{female}, \text{credit} = \text{bad}\}$ is jointly responsible for Alice's application being denied because there exists $\mathbf{c}' = \{\text{gender} = \text{male}, \text{credit} = \text{good}\}$ that would change the decision and $\mathbf{c}^*$ is minimal. But what can we say about the individual features? We could say that the whole set bears responsibility but individual features do not, or that any feature in a cause set is partially and equally accountable. Both approaches have issues, as they imply the features are equally responsible. This conflicts with our intuition: OBCB did not evaluate Alice's credit score, so her denial was caused by her gender.⁶

Prior work suggests a way out of this difficulty: hold some of the context in which counterfactuals are evaluated fixed, calling such fixed variables *witnesses*. Let us focus on the mediator *check-failed* between *gender* and *loan*. If we fix this variable at the factual value by intervening, then counterfactually, Alice would have been granted the loan if she were male. However, there is no context fixed at the actual value such that, had her credit score been higher, she would have been granted the loan. This motivates the following formalization:

Definition 4 (Actual cause). *Let $\mathbf{C} \subseteq \mathbf{S}$ be a cause set, $\mathbf{W} \subseteq \mathbf{V}$ be a set of possible witnesses. $\mathbf{C} = \mathbf{c}^*$ is causally responsible for $Y = y^*$ in $(M, \mathbf{u})$ just in case there is a $\mathbf{T}$ (an actual witness set to be held fixed) s.t. $\mathbf{T} \subseteq \mathbf{W}, \mathbf{T} \cap \mathbf{C} = \emptyset$ and $(M, \mathbf{u}) \models \mathbf{C} = \mathbf{c}^*, Y = y^*$ and $(M, \mathbf{u})[\mathbf{C} \leftarrow \mathbf{c}', \mathbf{T} \leftarrow \mathbf{t}^*] \not\models Y = y^*$, for some $\mathbf{c}'$ distinct everywhere from $\mathbf{c}^*$, where no proper subset of $\mathbf{C}$ satisfies this property.*

This definition extends but-for causality with a witness set $\mathbf{T}$ that is fixed to its factual value by an intervention. Under this definition, to determine the cause of an outcome, we need to find a cause set $\mathbf{C}$, alternative values $\mathbf{c}'$ (trivial if the variables are binary and less so if they are categorical or continuous), and a witness set $\mathbf{T}$. $\mathbf{C}$ and $\mathbf{T}$ are drawn from sets of size at most $2^{|\mathbf{V}|}$, so a brute-force computation of these sets is expensive.

Putting computational difficulties aside, we have so far considered deterministic models, but would like to capture probabilistic models as well, so we now consider a stochastic variant of the OBCB problem.

2.2 Probability of Necessity and Sufficiency

Example 5 (OBCB, stochastic). *The bank performs credit checks for a randomly selected 20% of women and skips checks for a randomly selected 10% of men. Unchecked applicants are rejected. Furthermore, if a man's credit check fails, he is nevertheless accepted 5% of the time and if a woman's credit check succeeds, she is denied 10% of the time. We assume that the population is evenly divided into male and female and good and bad credit. Formally, the model, using 1 for male and 1 for good credit, is:*

⁶Examples of this sort, involving both over-determination and undercutting of causal impact, abound. Suppose Jim is both stabbed and poisoned and dies as a result. Because stabbing works quickly, it kills Jim before the poison takes effect. In this case, it's the stabbing that caused his death. Yet analogous but-for clauses fail to break the symmetry here as well.

$loan-prob = \begin{matrix} gender=0 \\ gender=1 \end{matrix}$	$\begin{matrix} check-failed=0 & check-failed=1 \\ 0.9 & 0 \\ 1 & 0.05 \end{matrix}$
--	--

$$\begin{aligned}
credit &\sim \text{Bern}(0.5) \\
gender &\sim \text{Bern}(0.5) \\
check &\sim \text{Bern}((1 - gender) \cdot 0.2 + gender \cdot 0.9) \\
check-failed &= check \cdot (1 - credit) \\
loan-if-checked &\sim \text{Bern}(loan-prob[gender, check-failed]) \\
loan &= loan-if-checked \cdot check
\end{aligned}$$

The witness mechanism works well for deterministic models, but actual causality in its standard form offers no straightforward way to handle probabilistic ones. Let’s shelve that problem for now and instead ask: what would a probabilistic approach to causal attribution look like, and what would it miss without witnesses?

Before extending the witness idea to the stochastic setting, it is instructive to examine a probabilistic approach that scales more naturally: Pearl’s probabilities of necessity and sufficiency [Pearl, 2022]. This will let us identify what probabilistic machinery is needed, and what context-sensitivity is lost when witnesses are absent — motivating the synthesis that PCI provides. The following definitions allow for probabilistic models, but assume that all variables are Boolean (PCI will relax this assumption, allowing for continuous variables). Pearl considers two ways that a cause may relate to an effect: it may be necessary or sufficient for the effect. A cause is necessary if, counterfactually absent the cause, the effect is unlikely to occur; it is sufficient if the effect is counterfactually likely in the presence of the cause. We now formalize these ideas and discuss them in the context of the running example.

Definition 6 (Probability of Necessity). *Let the probability of necessity be the probability that the outcome $Y = y$ would not have occurred in the absence of $C = c^*$, given that both occurred:*

$$PN(C = c^*, Y = y^*) = P(Y_{c'} \neq y^* \mid C = c^*, Y = y^*),$$

where $Y_{c'}$ is the value of Y after intervening on C with $c' \neq c^*$.

Strictly speaking, the probability of necessity is less specific to a particular situation than the actual cause test discussed above. For example, we can consider the probability that being male causes someone to fail to get a loan, but we can’t consider the probability that being male causes Bob specifically (who we know has bad credit) to fail to get a loan. We might try to patch this up by conditioning, for the time being (while also reporting population-level values that follow the definitions more closely).

We illustrate the computation for Alice’s gender. Alice is factually $gender = \text{female}$, $credit = \text{bad}$, $loan = F$; the counterfactual intervenes to set $gender = \text{male}$ while leaving $credit = \text{bad}$. Tracing through the model:

- Under the intervention, $check \sim \text{Bern}(0.9)$ — men are checked 90% of the time.
- If checked, $check-failed = 1$ (Alice’s credit is bad), and from the loan-probability table $P(loan-if-checked = 1 \mid gender = \text{male}, check-failed = 1) = 0.05$.
- If not checked, $loan = 0$ deterministically since $loan = loan-if-checked \cdot check$.

Combining the two branches:

$$\begin{aligned}
PN(gender = \text{female}, loan = F \mid \text{Alice}) \\
&= P(loan_{gender=\text{male}} = T \mid gender = \text{female}, credit = \text{bad}) \\
&= 0.9 \cdot 0.05 = 0.045.
\end{aligned}$$

Computing the analogous quantities for the remaining (variable, person) pairs gives the individual-level values shown in Table 1; the table also lists population-level values for comparison.

The PN scores partially align with intuition: Bob’s bad credit is high ($PN = 0.90$), and his gender score is zero, matching the reading that his gender played no role. But Alice’s PN values do not separate gender (0.045) from credit (0.18) particularly well, and Alice’s gender PN is only marginally above Bob’s — not the sharp distinction we would want, given that the bank never even evaluated Alice’s credit. PN cannot register this asymmetry: it asks counterfactually what would happen under intervention without any context about which causal paths were active. For the full picture we need a second quantity, the probability of sufficiency.

Definition 7 (Probability of Sufficiency). *Let the probability of sufficiency be the probability that the outcome $Y = y^*$ would have occurred in the presence of $C = c^*$, given that neither $C = c^*$ nor $Y = y^*$ occurred:*

$$PS(C = c^*, Y = y^*) = P(Y_{c^*} = y^* \mid C = c', Y \neq y^*)$$

where $c' \neq c^*$ and Y_{c^*} is the value of Y after intervening on C with c^* .

To query whether the factual cause was sufficient, we condition on the counterfactual: we assume the cause did *not* take its factual value and the outcome did *not* occur, then intervene to restore the factual cause value and ask whether the outcome now occurs. Intuitively, this asks: if the cause had been absent and things had gone differently, would reinstating the cause have brought the outcome about? Let’s look at the resulting values.

We illustrate PS for Alice’s gender. Following the same convention as for PN , the individual-level computation additionally conditions on Alice’s credit ($credit = bad$). The PS conditional then has Alice’s credit fixed but the cause and outcome flipped to their non-factual values ($gender = male$, $loan = T$); we intervene to restore $gender = female$ and ask whether the loan is again denied:

- With $gender = female$ and $credit = bad$, $check \sim \text{Bern}(0.2)$.
- If checked, $check-failed = 1$, and from the loan-probability table $P(loan-if-checked = 1 \mid gender = female, check-failed = 1) = 0$.
- If not checked, $loan = 0$ deterministically.

Either branch gives $loan = F$, so

$$\begin{aligned} PS(gender = female, loan = F \mid \text{Alice}) \\ = P(loan_{gender=female} = F \mid gender = male, loan = T, credit = bad) = 1. \end{aligned}$$

The remaining individual values, along with population-level values for comparison, appear in Table 1. Two cases warrant comment: Bob’s gender PS is undefined because the required conditioning event ($gender = female$, $credit = bad$, $loan = T$) has probability zero in the model; Bob’s credit PS is 0.955 rather than 1 because the model includes a 5% exception rate for males who fail the credit check.

To somehow combine insights provided by the two scores, we would like to find causes that are both necessary and sufficient. This notion is formalized as follows:

Definition 8 (Probability of necessity and sufficiency). *Let the probability of necessity and sufficiency be the probability that the outcome $Y = y^*$ would have occurred in the presence of $C = c^*$ and that it would not have occurred under the intervention $C = c'$:*

$$PNS(C = c^*, Y = y^*) = P(Y_{c^*} = y^*, Y_{c'} \neq y^*)$$

The population- and individual-level PNS values appear in the rightmost column group of Table 1. These values capture the relative causal impact of $gender$ and $credit$ but are not entirely satisfying. Table 1 reveals the problem directly: for Alice, $PNS(gender) = 0.045 < PNS(credit) = 0.18$, so PNS ranks credit as more causally responsible than gender. Yet the bank never evaluated Alice’s credit—the $check$ variable was F throughout. PNS has no mechanism to register this: it asks counterfactually what would happen if credit were different, without fixing $check-failed$, the mediating variable that blocked credit’s causal path.

As an attempt to help Alice better understand her particular situation, we might consider a modification of PNS with additional conditioning:

$$PNS_c(C = c^*, Y = y^* \mid F = f^*) = P(Y_{c^*} = y^*, Y_{c'} \neq y^* \mid F = f^*)$$

Using this definition, we can compute $PNS_c(\text{gender} = \text{female}, \text{loan} = F \mid \text{credit} = \text{bad}) = 0.045$ and $PNS_c(\text{credit} = \text{bad}, \text{loan} = F \mid \text{gender} = \text{female}) = 0.18$. Conditioning does not solve the problem described above—Alice’s credit was in reality never evaluated—because the interventions override the mediating variable that tracks whether her credit has been checked. When we compute $PNS_c(\text{credit} = \text{bad}, \text{loan} = F \mid \text{gender} = \text{female})$, conditioning on gender being female is consistent with the factual—but when we then intervene to set credit to good, this intervention propagates downstream and changes *check-failed*, overriding what we learned from conditioning on Alice’s factual outcome. The context we fixed is undone by the very intervention meant to probe it. We will not consider this use of conditioning further, and this observation motivates the need for intervention-based (as opposed to conditioning-based) context-sensitivity, which is what the witness mechanism provides.

Table 1: PN , PS , and PNS for Alice and Bob in Example 5, with $\text{loan} = F$ as the outcome. “Population” values condition only on the cause–outcome pair prescribed by the definition; “Individual” values additionally condition on the subject’s other factual variable (e.g. for Alice’s gender, on $\text{credit} = \text{bad}$). A dash marks an entry that is undefined because the conditioning premises are inconsistent with the model.

Variable	Level	Alice			Bob		
		PN	PS	PNS	PN	PS	PNS
<i>gender</i>	Population	0.43	0.83	0.39	0.02	0.10	0.01
<i>gender</i>	Individual	0.045	1.00	0.045	0	—	0
<i>credit</i>	Population	0.53	0.96	0.52	0.53	0.96	0.52
<i>credit</i>	Individual	0.18	1.00	0.18	0.90	0.955	0.86

2.3 Path forward

What should we conclude from this investigation of actual causality and the probability of necessity and sufficiency? First, we need a way to treat probabilistic models, and the probability of necessity and sufficiency gives us a tool for doing that. However, PNS is not as specific as we would like because it does not capture relevant features of individual units that are not part of the cause. Actual causality offers more specificity, but requires modification to deal with probability and continuous variables, and is computationally expensive to compute.

PCI addresses both limitations by combining the two approaches. To see the intuition, consider Alice’s case once more. When PCI evaluates gender’s causal impact on her denial, it considers counterfactual worlds in which her gender is changed, while drawing witness sets from a distribution that includes *check-failed* at its factual value of F — the credit check was never performed. In those worlds, changing gender to male, with *check-failed* held fixed at F (by intervention, so unlike in the case of patched-up PNS, the mediator value does not get overridden), leads to loan approval: a large counterfactual shift, and so high attribution. When PCI evaluates credit’s causal impact, it searches for witness sets that, held fixed, would make a change to good credit matter for Alice — but no such context exists, because the credit check was never run. The distribution over witness sets therefore assigns low weight to any world in which credit is relevant for Alice, and her credit receives low attribution. Unlike actual causality, PCI does not ask whether *some* witness set validates a counterfactual dependency; it integrates over the distribution of all witness sets, weighting each by its probability. Unlike PNS, it inherits the context-sensitivity of actual causality through this integration. The formal definitions in Section 3 make this expectation precise — but the intuition is the one we have just seen: PCI inherits the witness mechanism of actual causality and the probabilistic language of PN/PNS, and generalizes and combines them into a tractable estimation problem even when variables are continuous.

3 Causal Impact: definitions

Explanation and causal attribution seek to connect a potentially causal *event* $X = x$ to an outcome event $Y = y$.⁷ In this section, we present a formal development. To ground it, we return to Example 5 and use Alice and Bob as running illustrations throughout this section. Both are rejected for a loan; the question is which of their attributes—*gender* and *credit*—was causally responsible for each rejection, and to what degree. While the example is discrete, the machinery applies to continuous and mixed-type variables; a continuous illustration is given in Section ??.

Add cross-reference to continuous example.

Let $R(V \rightsquigarrow \text{loan} \mid \text{person})$ denote the causal responsibility of variable V for the loan outcome at the individual level (R is a placeholder for any attribution score and should not be identified with a specific formal quantity; PCI interprets R as $\mathbb{E}[ci(Y^s, Y^n, y^*)]$, to be defined below, but the desiderata are method-agnostic). Attribution methods may produce signed values; the desiderata are stated in terms of absolute magnitudes $|R(\cdot)|$, allowing for directionality while remaining agnostic about sign conventions. The following desiderata encode what an intuitively correct causal responsibility attribution should look like:

- D-A1** $|R(\text{gender} \rightsquigarrow \text{loan} \mid \text{Alice})| > 0$
- D-A2** $|R(\text{credit} \rightsquigarrow \text{loan} \mid \text{Alice})| > 0$
- D-A-rank** $|R(\text{gender} \rightsquigarrow \text{loan} \mid \text{Alice})| > |R(\text{credit} \rightsquigarrow \text{loan} \mid \text{Alice})|$
- D-B1** $|R(\text{gender} \rightsquigarrow \text{loan} \mid \text{Bob})| > 0$
- D-B2** $|R(\text{credit} \rightsquigarrow \text{loan} \mid \text{Bob})| > 0$
- D-B-rank** $|R(\text{credit} \rightsquigarrow \text{loan} \mid \text{Bob})| > |R(\text{gender} \rightsquigarrow \text{loan} \mid \text{Bob})|$
- D-comp** $|R(\text{gender} \rightsquigarrow \text{loan} \mid \text{Alice})| > |R(\text{gender} \rightsquigarrow \text{loan} \mid \text{Bob})|$

D-A1 and **D-A2** both require non-zero attribution for Alice because two distinct causal routes were available: *gender* filtered her out at the checking stage without her *credit* ever being evaluated, while bad *credit* would have mattered had she been checked. Neither is causally inert. **D-A-rank** is the substantive constraint: because Alice never underwent a credit evaluation, *gender*’s causal role is immediate and structural, whereas *credit*’s is purely hypothetical. A method that ranks *credit* above *gender* for Alice is misattributing the cause of her rejection.

For Bob, **D-B2** reflects that the credit check occurred and directly produced the rejection: the causal chain $\text{credit} \rightarrow \text{check-failed} \rightarrow \text{loan}$ was activated. **D-B1** requires strictly positive attribution for *gender*: being male gave Bob a 90% probability of being checked, so *gender* opened the evaluation that rejected him. This indirect causal role is non-zero; a method returning exactly zero fails to capture it. **D-B-rank** corresponds to the intuition that *credit* is the proximate cause. **D-comp** captures the cross-individual asymmetry that is hardest for naive methods to express: *gender* is more causally responsible for Alice’s rejection than for Bob’s, because for Alice *gender*=female was sufficient on its own to produce the rejection — no evaluation took place, so *credit* was never a factor — while for Bob *gender*=male merely opened the evaluation that *credit*=bad then caused. We will verify at the end of this section that PCI satisfies all seven desiderata, while PNS fails **D-A-rank** and **D-B1**.

We work within the context of probabilistic structural causal models [Pearl, 2009].

Definition 9 (Probabilistic Structural Causal Model (PSCM)). *A probabilistic structural causal model is a tuple*

$$M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle,$$

where $\mathbf{V} = \{V_1, \dots, V_n\}$ is a finite set of endogenous variables, $\mathbf{U} = \{U_1, \dots, U_n\}$ is a set of exogenous variables, and $\mathbf{F} = \{f_1, \dots, f_n\}$ is a set of structural assignments

$$V_i := f_i(\mathbf{Pa}_i, U_i), \quad i = 1, \dots, n,$$

⁷This is quite different from type-level causal queries, where one is interested in some group-level attribution, as when one estimates average treatment effect or conditional average treatment effect.

with $\text{Pa}_i \subseteq \mathbf{V} \setminus \{V_i\}$ denoting the parents of V_i . The distribution $P_{\mathbf{U}}$ is a probability measure on $\text{dom}(\mathbf{U})$, and the assignments in $\mathbf{F}$ are assumed to be acyclic and to have a unique solution for $\mathbf{V}$ given $\mathbf{U}$.⁸

A PSCM encodes which variables influence which others, through what mechanisms, and subject to what randomness. The structural equations specify each variable as a deterministic function of its causal parents and local noise; the exogenous distribution captures all irreducible uncertainty in the system. Together they make every counterfactual question well-posed and computable.

The OBCB model (Example 5) is a PSCM with endogenous variables

$$\mathbf{V} = \{\text{gender}, \text{credit}, \text{check}, \text{check-failed}, \text{loan}\},$$

and exogenous variables

$$\mathbf{U} = \{U_{\text{gender}}, U_{\text{credit}}, U_{\text{check}}, U_{\text{loan}}\}$$

which induce independent Bernoulli noise.

The structural assignments in the case at hand can be written as:

$$\begin{aligned} \text{gender} &\sim \text{Bernoulli}(0.5), & \text{credit} &\sim \text{Bernoulli}(0.5), \\ \text{check} &\sim \text{Bernoulli}(0.2(1 - \text{gender}) + 0.9\text{gender}), \\ \text{check-failed} &:= \text{check}(1 - \text{credit}), \\ \text{loan} &\sim \text{Bernoulli}(\text{loan-prob}[\text{gender}, \text{check-failed}]) \cdot \text{check}. \end{aligned}$$

Definition 10 (Interventions). *Given a model $M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$, an intervention on variables $\mathbf{X} = \{X_1, \dots, X_k\} \subseteq \mathbf{V}$ with values $\mathbf{x} = (x_1, \dots, x_k)$ replaces the structural assignments for X_i by $X_i := x_i$, where $i = 1, \dots, k$. We denote this intervention by either $[\mathbf{X} \leftarrow \mathbf{x}]$ or $\text{do}(\mathbf{X} = \mathbf{x})$.*

An intervention surgically replaces a variable’s structural equation with a fixed constant, disconnecting it from its usual causes. The rest of the model propagates downstream as normal under the new assignment; the exogenous noise is unchanged.

In the OBCB model, $\text{do}(\text{gender} = 1)$ replaces Alice’s gender assignment with male, leaving all other structural equations intact. The downstream effect propagates: *check* now draws from $\text{Bern}(0.9)$ rather than $\text{Bern}(0.2)$, so Alice would be checked in 90% of noise realizations instead of 20%. This is the counterfactual world in which we ask whether her loan outcome would have differed.

Because PCI supports arbitrary variable types, it will be helpful to define a measure theoretic object corresponding to interventional outcomes. Because outcomes are a deterministic function of exogenous noise, this will be a Dirac measure.

Definition 11 (Interventional Law). *Let $M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$ be a probabilistic structural causal model and let $Y \in \mathbf{V}$. For an intervention $\text{do}(\mathbf{X} = \mathbf{x})$, let $Y' : \text{dom}(\mathbf{U}) \rightarrow \text{dom}(Y)$ denote the measurable map induced by the modified structural assignments. The interventional law of Y given fixed $\mathbf{u} \in \text{dom}(\mathbf{U})$ is the probability measure on $\text{dom}(Y)$ defined by*

$$P(Y \in A \mid \mathbf{u}, \text{do}(\mathbf{X} = \mathbf{x})) := \mathbb{I}\{Y'(\mathbf{u}) \in A\} = \delta_{Y'(\mathbf{u})}(A), \quad A \subseteq \text{dom}(Y).$$

The interventional law records the deterministic outcome of Y under $\text{do}(\mathbf{X} = \mathbf{x})$ for a fixed noise realization $\mathbf{u}$: since the modified structural equations make Y a function of $\mathbf{u}$ alone, its distribution collapses to a point mass. Integrating over $P_{\mathbf{U}}$ then recovers the full interventional distribution.

We define a *causal set* to be a set of assignments to the variables of a probabilistic structural causal model.

Definition 12 (Causal set). *Given a model $\langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$, a causal set is a set of pairs $X = x$ where $X \in \mathbf{V}$ and $x \in \text{dom}(X)$.*

⁸Working within the PSCM framework commits the practitioner to fully specified structural assignments $\mathbf{F}$ and an exogenous distribution $P_{\mathbf{U}}$ — analogous to how Bayesian inference requires a likelihood and a prior. This is precisely what enables PCI to yield individual-level counterfactual attributions rather than population-level bounds, and once $\mathbf{F}$ and $P_{\mathbf{U}}$ are specified, sampling from $P_{\mathbf{U}}$ is a routine probabilistic inference operation.

Informally, a causal set is a list of specific variable-value assignments — a precise statement of which variables are set to which values. It is the basic unit of intervention: from a causal set one can read off exactly what is being manipulated or held fixed.

In PCI, causal sets play four distinct roles — two on the cause side and two on the context side:

Suspects $\mathbf{S}$. The full set of variables suspected to bear on the outcome. Subsets of $\mathbf{S}$ will be sampled and intervened on.

Active suspects $\mathbf{C} \subseteq \mathbf{S}$. The specific subset intervened on in a given search step — either to alternative values (necessity regime) or to factual values (sufficiency regime).

Witnesses $\mathbf{W}$. The full set of context variables that may be held fixed at their factual values. Subsets of $\mathbf{W}$ will be sampled and held fixed.

Active witnesses $\mathbf{T} \subseteq \mathbf{W}$. The specific subset held fixed in a given search step. To avoid contradicting the active intervention, $\mathbf{T}$ is pruned so that $\mathbf{T} \cap \mathbf{C} = \emptyset$.

To evaluate the causal impact of a variable on an outcome, we consider two interventional regimes: the *necessity regime*, in which we intervene on the variables $\mathbf{C} \subseteq \mathbf{S}$ to have an alternative value, and the *sufficiency regime*, in which we intervene so that $\mathbf{C}$ takes on factual values. Moreover, we hold some elements $\mathbf{T} \subseteq \mathbf{W}$ of the context fixed (i.e., intervened on to retain factual values, despite intervention on $\mathbf{C}$). When looking for an explanation, we perform a search across possible cause sets and possible contexts. Complete model samples in these regimes will be called *necessity worlds* and *sufficiency worlds*, respectively.

The intuition is that the suspect set $\mathbf{S}$ names every variable that could plausibly have caused the outcome; the witness set $\mathbf{W}$ names context variables to be selectively held fixed, isolating the causal channels under investigation. Intervening on suspects tests what would have happened under an alternative cause in the necessity worlds and under the factual values in the sufficiency worlds; fixing witnesses blocks confounding paths that would otherwise absorb the effect. Responsibility attribution increases when the outcome is far from the factual value in the necessity world and decreases when it is far from the factual value in the sufficiency world.

For both Alice (female, bad credit, loan denied) and Bob (male, bad credit, loan denied) we set $\mathbf{S} = \{\text{gender}, \text{credit}\}$ and $\mathbf{W} = \{\text{check-failed}\}$. The witness *check-failed* records whether the applicant was checked and had bad credit; holding it at its factual value isolates the causal path from *gender* or *credit* to *loan* independently of the credit-check bottleneck. Without this witness, intervening on *credit* for Alice would leave 80% of noise realizations unaffected — those in which she is never checked — making credit’s causal role invisible. The witness makes the bottleneck explicit and manipulable.

Moving on with the general framework, we will be searching for causes by intervening on subsets of suspects and holding fixed subsets of witnesses. To make this search tractable, we need a way to prioritize which subsets to consider.

Definition 13 (Variable Selection Distribution). *Let $\mathbf{X}$ be a finite set of random variables. A variable selection distribution Γ is a probability distribution on $2^{\mathbf{X}}$. We write $\mathbf{Y} \sim \Gamma$ to mean $\mathbf{Y}$ is a random subset of $\mathbf{X}$ drawn from Γ , with probability mass function $p^\Gamma(\mathbf{y}) = \mathbb{P}(\mathbf{Y} = \mathbf{y})$ for $\mathbf{y} \subseteq \mathbf{X}$.*

The variable selection distribution Γ encodes prior beliefs about which subsets of suspects are worth examining. A uniform distribution treats all suspect subsets as equally plausible; a cardinality-constrained choice focuses the search on interactions of a particular order, analogous to choosing which interaction terms to include in a regression. More complex distributions could encode domain knowledge about which variables are more likely to interact, or could be learned from data. The choice of Γ is a design decision that shapes the search process and the resulting attributions. As a default, a uniform Γ over the full powerset is the least-informative starting point, assigning equal weight to all subset sizes, analogous to including all interaction orders in a regression. There is no universally principled criterion for restricting the cardinality range; the honest advice is to treat J and K as explicit design choices. Two practical considerations bear on this choice: with $K = 1$, attribution scores are based on singleton interventions and straightforward to communicate to stakeholders; higher K averages over joint interventions on X_k with other suspects simultaneously, which is harder to explain and, in large or structurally complex models, may amplify estimation noise rather than capturing genuine higher-order causal structure. For applications where attribution

has serious consequences, we additionally recommend a cardinality sensitivity analysis: if rankings are stable as K increases from 1 to $|\mathbf{S}|$, the simpler choice is supported; if they diverge, higher-order interactions are materially relevant and the full powerset should be used. The key advantage of PCI is that this commitment is made transparently rather than absorbed into an implicit default.

In particular, we will be using variable selection distributions for the powerset of causal suspects $\mathbf{S}$ and for the powerset of witnesses $\mathbf{W}$. One simple pattern of suspect and witness selection can be derived by setting size limits on the number of activated suspects or witnesses, and sampling uniformly otherwise.

Example 14 (Cardinality-Constrained Uniform Selection). *Let $\mathbf{X}$ be a candidate set of random variables and let $k \sim \text{Unif}\{J, \dots, K\}$ with $1 \leq J \leq K \leq |\mathbf{X}|$. A variable selection $\mathbf{Y} \sim \Gamma(2^{\mathbf{X}})$ is sampled from a cardinality-constrained uniform selection distribution $\Gamma(2^{\mathbf{X}})$ with probability mass function*

$$p^\Gamma(\mathbf{Y}) \propto \mathbb{I}[|\mathbf{Y}| = k].$$

*The parameters J and K determine the cardinality range of the search; specific choices connect PCI to existing methods and enable efficient Monte Carlo approximation.*⁹

In PCI the variable selection distribution Γ is a probability mass function on $2^{\mathbf{S}} \times 2^{\mathbf{W}}$, governing the joint choice of an active suspect set $\mathbf{C} \subseteq \mathbf{S}$ and an active witness set $\mathbf{T} \subseteq \mathbf{W}$. The recommended construction samples antecedents and witnesses separately and then enforces the constraint that no variable plays both roles in the same draw: pick a marginal Γ_s on $2^{\mathbf{S}}$ and a marginal Γ_w on $2^{\mathbf{W}}$, draw $\mathbf{C} \sim \Gamma_s$ and $\mathbf{T} \sim \Gamma_w$ independently, and *reject* any draw with $\mathbf{T} \cap \mathbf{C} \neq \emptyset$, resampling until the constraint holds. Equivalently,

$$p^\Gamma(\mathbf{C}, \mathbf{T}) = \frac{p_s^\Gamma(\mathbf{C}) p_w^\Gamma(\mathbf{T}) \mathbb{I}[\mathbf{T} \cap \mathbf{C} = \emptyset]}{Z}, \quad Z = \sum_{(\mathbf{C}, \mathbf{T})} p_s^\Gamma(\mathbf{C}) p_w^\Gamma(\mathbf{T}) \mathbb{I}[\mathbf{T} \cap \mathbf{C} = \emptyset].$$

The normalizer Z has a closed-form expression: conditioning on $\mathbf{T}$, $Z = \mathbb{E}_{\mathbf{T} \sim \Gamma_w}[\text{Pr}_{\mathbf{C} \sim \Gamma_s}[\mathbf{C} \subseteq \mathbf{S} \setminus \mathbf{T}]]$, a finite sum over $2^{|\mathbf{W}|}$ terms (and analogously by conditioning on $\mathbf{C}$); no Monte Carlo is needed to evaluate it. Throughout, Γ_s and Γ_w are abbreviations for the sampling marginals of this construction; the formal primitive in Definition 17 is the joint Γ .

For the OBCB analysis, with $|\mathbf{S}| = 2$, we use a Γ built from Γ_s uniform over the three non-empty subsets of $\mathbf{S}$ ($\{gender\}$, $\{credit\}$, $\{gender, credit\}$, each with probability $1/3$) and Γ_w uniform over $\{\emptyset, \{check-failed\}\}$, each with probability $1/2$. Since no element appears in both $\mathbf{S}$ and $\mathbf{W}$, the rejection step is vacuous, $Z = 1$, and Γ reduces to the product of marginals. In larger suspect sets the choice of cardinality range becomes substantive: sampling uniformly from the full powerset yields an expected intervention size of $|\mathbf{S}|/2$, which spreads attribution across large coalitions and may make individual feature contributions harder to isolate; we return to this design decision in the experimental section.

⁹When $J = 1$ and $K = |\mathbf{X}|$, this distribution coincides with the Shapley weighting kernel, which samples coalition size uniformly and then selects a coalition uniformly at that size [Shapley, 1953]. The parameters J and K allow the practitioner to restrict the search to cause sets of a specific cardinality range: setting $J = K = 1$ confines attribution to individual variables, while $J = 2$, $K = 3$ focuses on small multi-variable interactions. Moreover, whereas exact Shapley computation requires summing over all $2^{|\mathbf{X}|}$ coalitions, PCI treats Γ as a generative model: approximation proceeds by sampling subsets from Γ and outcomes from P_U , making Monte Carlo estimation a native feature of the framework rather than an external algorithmic layer. We note that the weighting alone does not distinguish PCI from SHAP at $J = 1$, $K = |\mathbf{X}|$: the subset weights are identical. The distinction lies in the integrand, and it matters for what is being evaluated and what results follow. SHAP averages marginal contributions using the observational conditional $\mathbb{E}[f(\mathbf{X}_S = \mathbf{x}_S, \mathbf{X}_{\bar{S}})]$, marginalising over the empirical distribution of non-suspect features; this means a variable that is merely correlated with a cause — but not itself causally active — can receive positive attribution. PCI instead evaluates counterfactual outcomes $P(Y \in A \mid \mathbf{u}, \text{do}(\mathbf{C} = \mathbf{c}', \mathbf{T} = \mathbf{t}^*))$ computed via the structural model, so attribution flows only along genuine causal paths. The witness mechanism adds a further distinction: by holding intermediate variables fixed, PCI can resolve overdetermination and undercutting scenarios where SHAP assigns equal or incorrect attribution. The choice of J and K remains a design decision; practitioners can set $J = K = 1$ for single-variable attribution or allow higher-order interactions as discussed above. A detailed comparison of PCI, SHAP, and Causal SHAP on concrete examples is given in Section 4.

Causal attribution hinges on questions such as “if X had, contrary to fact, been fixed to $x' \neq x$ instead of x , what would the outcome have been?” For non-binary causes, this is ambiguous. To resolve this in a way that does not require committing to a single, contrastive causal event [Kawakami et al., 2024], we rely on a *distribution* over alternative values.

Definition 15 (Alternative Value Distribution). *Let $\mathbf{X}$ be a set of random variables with domain $\text{dom}(\mathbf{X})$ and let $\mathbf{X} = \mathbf{x} \in \text{dom}(\mathbf{X})$ be a factual event. An alternative value distribution $\Delta(\mathbf{x})$ is a probability measure on $\text{dom}(\mathbf{X})$ such that $\mathbf{x} \notin \text{supp}(\Delta(\mathbf{x}))$.*

Informally, $\Delta(\mathbf{x})$ specifies what counts as a genuinely different value for a suspect variable. The support condition ensures no mass falls on the factual value $\mathbf{x}$ itself, so every sampled alternative is a real counterfactual departure rather than a noisy copy of what actually happened.

When $\Delta(\mathbf{x})$ is a posterior conditioned on upstream factual values — the construction we use throughout — alternatives are weighted by their plausibility under the structural model with upstream context held fixed. The upstream-only conditioning is itself a normality-respecting choice: factual upstream values play the role of the *default* against which counterfactual alternatives are graded, while downstream evidence (which would leak the consequence back into the counterfactual) is deliberately excluded. This is the operational link to the descriptive-normality literature [Halpern and Hitchcock, 2015, Icard et al., 2017] mentioned in the introduction; PCI does not aim to resolve the counterexamples motivating that line of work, only to share its core intuition that causal attribution depends on what counts as a plausible counterfactual.

One natural alternative value distribution results from taking the one currently embodied in the model and excising it around the factual setting. Then, the semantics for “alternative value” means “a sufficiently different value, following the original distribution otherwise”. In the discrete case, of course, the analogue is simply the probability mass function conditioned on the factual value not holding.

Example 16 (ϵ -Excised Distribution). *Assume $\mathbf{X}$ is $\mathbb{R}^d$ -valued with density p . Given a factual event $\mathbf{X} = \mathbf{x}$, define the ϵ -excised alternative value distribution $\Delta_\epsilon(\mathbf{x})$ as the probability measure on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ with density*

$$p_{\Delta_\epsilon}(\mathbf{x}' | \mathbf{x}) \propto \begin{cases} 0, & \mathbf{x}' \in B_\epsilon(\mathbf{x}), \\ p(\mathbf{x}'), & \text{otherwise,} \end{cases}$$

where $B_\epsilon(\mathbf{x})$ denotes the ϵ -ball centered at $\mathbf{x}$. In particular, $\mathbf{x} \notin \text{supp}(\Delta_\epsilon(\mathbf{x}))$.

Informally, the alternative value distribution Δ operationalises what “counterfactually different” means for each suspect variable: it specifies which alternative values are considered when asking whether the outcome would have changed under an alternative cause.

In the OCB model both *gender* and *credit* are binary, so Δ is trivial. For Alice, the only alternative to *gender* = 0 (female) is *gender* = 1 (male), and the only alternative to *credit* = 0 (bad) is *credit* = 1 (good); for Bob, the only alternative to *gender* = 1 is *gender* = 0. Accordingly, $\Delta(\text{gender} = 0)$ is a point mass on *gender* = 1 and $\Delta(\text{credit} = 0)$ a point mass on *credit* = 1 — no distributional choice is required. The binary case is exactly where Δ is unambiguous; it is for continuous suspects such as income or loan amount that the ϵ -excised distribution becomes essential, forcing an explicit commitment to what counts as a meaningfully different value.¹⁰

¹⁰Any method that asks “what would happen if $\mathbf{X}$ had been different” must operationalise what *different* means. Using the empirical distribution $p(\mathbf{x}')$ directly — as in kernel SHAP [Lundberg and Lee, 2017] — is not a neutral choice: because p places mass near the factual value $\mathbf{x}$, sampled alternatives may be practically indistinguishable from $\mathbf{x}$, conflating the necessity question (“what if $\mathbf{x}$ had genuinely differed?”) with evaluating essentially the same setting. More generally, three natural choices exist for Δ . The observational marginal $p(\mathbf{x}'_k | \mathbf{x}_{-k})$ conflates the necessity question with the factual setting, as noted above. The fully conditional posterior $p(\mathbf{x}'_k | \mathbf{x}_{-k})$ conditions on all other observed features — potentially including descendants of X_k — which imposes structural constraints that may rule out otherwise plausible counterfactuals, and still concentrates mass near x_k when features are correlated. The ϵ -excised distribution occupies the middle ground: it draws from the marginal informed by upstream variables only, excising the factual neighbourhood to guarantee genuine departure without downstream conditioning. The ϵ -excised distribution makes the commitment explicit and tunable: *different* means outside $B_\epsilon(\mathbf{x})$, with ϵ set by the practitioner to reflect the minimum change deemed mean-

We are now ready to state the main definitions of PCI. The core idea is to measure two complementary things for a variable of interest X_k : (a) how much X_k 's factual value contributes to producing y^* — the *sufficiency* question — and (b) how much replacing X_k 's value with alternatives would shift the outcome away from y^* — the *necessity* question. We capture both by constructing two counterfactual outcome distributions conditioned on the same noise realization $\mathbf{u}$, then marginalising their product over $P_{\mathbf{U}}$. Because PCI is not constrained to particular variable types, we work in general measure-theoretic terms supporting combinations of continuous and discrete variables.

Definition 17 (Joint Necessity and Sufficiency Measure). *Let $M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$ be a probabilistic structural causal model and Y an outcome variable with domain $\text{dom}(Y)$. Let p^Γ be the mass function of a variable selection distribution on $2^{\mathbf{S}} \times 2^{\mathbf{W}}$, and let $\Delta(\mathbf{s}^*)$ be an alternative value distribution on $\text{dom}(\mathbf{S})$. Let $(\mathbf{S}, \mathbf{W}) = (\mathbf{s}^*, \mathbf{w}^*)$ be a factual event.*

For a variable of interest X_k , let $2_k^{\mathbf{S}} := \{\mathbf{C} \subseteq \mathbf{S} : X_k \in \mathbf{C}\}$, and for each $\mathbf{C} \subseteq \mathbf{S}$ let $\Delta_{\mathbf{C}}(\mathbf{s}^)$ denote the pushforward of $\Delta(\mathbf{s}^*)$ under the restriction map $\mathbf{s}' \mapsto \mathbf{s}'|_{\mathbf{C}}$.¹¹ For any measurable $A \subseteq \text{dom}(Y)$ and exogenous realization $\mathbf{u}$, the sufficiency and necessity measures are defined as follows. The sufficiency measure $P_k^s(A | \mathbf{u})$ intervenes on each active suspect set $\mathbf{C}$ to its factual values $\mathbf{s}^*|_{\mathbf{C}}$, holding active witnesses $\mathbf{T}$ fixed, and computes the Γ -weighted sum of $P(Y \in A | \mathbf{u}, \text{do}(\cdot))$ under factual cause restoration. The necessity measure $P_k^n(A | \mathbf{u})$ instead integrates over alternative values $\mathbf{c}'$ for $\mathbf{C}$, again holding $\mathbf{T}$ fixed at factual values, and computes the Γ -weighted average of $P(Y \in A | \mathbf{u}, \text{do}(\cdot))$ over alternative cause values.*

$$P_k^s(A | \mathbf{u}) = \sum_{\substack{(\mathbf{C}, \mathbf{T}) \\ X_k \in \mathbf{C}}} P(Y \in A | \mathbf{u}, \text{do}(\mathbf{C} = \mathbf{s}^*|_{\mathbf{C}}, \mathbf{T} = \mathbf{w}^*|_{\mathbf{T}})) p^\Gamma(\mathbf{C}, \mathbf{T}), \quad (1)$$

$$P_k^n(A | \mathbf{u}) = \sum_{\substack{(\mathbf{C}, \mathbf{T}) \\ X_k \in \mathbf{C}}} \int P(Y \in A | \mathbf{u}, \text{do}(\mathbf{C} = \mathbf{c}', \mathbf{T} = \mathbf{w}^*|_{\mathbf{T}})) \Delta_{\mathbf{C}}(\mathbf{s}^*)(d\mathbf{c}') p^\Gamma(\mathbf{C}, \mathbf{T}). \quad (2)$$

The joint necessity-sufficiency measure then is given by

$$P_k^{s,n}(A \times B) := \int P_k^s(A | \mathbf{u}) P_k^n(B | \mathbf{u}) P_{\mathbf{U}}(d\mathbf{u}). \quad (3)$$

Informally: $P_k^s(\cdot | \mathbf{u})$ is the distribution of Y in the *sufficiency world* — restore the factual value of a candidate cause set $\mathbf{C}$ containing X_k , hold the witness set $\mathbf{T}$ at its factual values, and observe where the outcome lands. $P_k^n(\cdot | \mathbf{u})$ is the distribution of Y in the *necessity world* — replace $\mathbf{C}$'s values with alternatives drawn from Δ , hold $\mathbf{T}$ fixed, and ask how the outcome shifts. The sum over suspect-containing pairs $\{(\mathbf{C}, \mathbf{T}) : X_k \in \mathbf{C}\}$ weights each configuration by its Γ -mass — more focused interventions receive whatever weight the practitioner assigns them; configurations with $X_k \notin \mathbf{C}$ would contribute 0 to the attribution and are absent from the sum, so P_k^s and P_k^n are sub-probability measures with total mass $\Pr_{\Gamma}[X_k \in \mathbf{C}] \leq 1$ rather than renormalized expectations. In the OBCB example, for $X_k = \text{gender}$ under a noise realization in which Alice is credit-checked: the sufficiency world restores *gender* = female while holding *check-failed* fixed; the necessity world substitutes *gender* = male under the same noise.

ingful. This parallels the Region of Practical Equivalence (ROPE) in Bayesian hypothesis testing [Kruschke, 2018]: just as ROPE forces an explicit decision about what difference is large enough to matter, ϵ -excision forces an explicit decision about what counterfactual is sufficiently far from the factual to count as genuinely alternative. The practitioner makes this commitment regardless of method; the only question is whether it is made transparently or absorbed into a default. For practical guidance on choosing ϵ : when X is standardised to $\mathcal{N}(0, 1)$, the choice acquires a direct probabilistic interpretation — the excised mass $\Phi(x^* + \epsilon) - \Phi(x^* - \epsilon)$ is the fraction of the distribution ruled out as insufficiently different from the factual value x^* . At $x^* = 0$, setting $\epsilon = 0.2$ excises approximately 15.9% of the distribution, $\epsilon = 0.5$ excises 38.3%, and $\epsilon = 0.8$ excises 57.6%, corresponding roughly to small, medium, and large effect thresholds in the sense of Cohen [1988]. Alternatively, ϵ can reflect measurement precision (the smallest meaningful increment of X) or a decision-relevant threshold (the minimum change in X that would alter a downstream outcome). As with ROPE, a sensitivity analysis across a range of ϵ values is advisable when the appropriate scale is unclear. For concrete examples of how SHAP's implicit choice leads to problematic attributions, see Section 4.

¹¹For an assignment $\mathbf{x}$ to a collection of variables $\mathbf{X}$ and any subset $\mathbf{Y} \subseteq \mathbf{X}$, the restriction $\mathbf{x}|_{\mathbf{Y}}$ denotes the sub-assignment of $\mathbf{x}$ to the variables in $\mathbf{Y}$.

The product decomposition in $P_k^{s,n}$ is legitimate because, conditional on $\mathbf{u}$, the sufficiency and necessity interventions operate in separate counterfactual regimes with no causal connection: $\text{do}(\mathbf{C} = \mathbf{c}^*)$ and $\text{do}(\mathbf{C} = \mathbf{c}')$ act on the same noise but in distinct hypothetical worlds, so Y^s and Y^n are conditionally independent given $\mathbf{u}$, and $P(Y^s \in A, Y^n \in B \mid \mathbf{u}) = P_k^s(A \mid \mathbf{u}) P_k^n(B \mid \mathbf{u})$.

With the necessity-sufficiency measure fully defined, we are ready to characterize the causal impact of an event X_k . In particular, we will proceed by defining an arbitrary *causal impact function*, whose expectation over the necessity-sufficiency measure determines the causal impact. We first proceed with a formal definition of the causal impact measure, followed by several intuitive candidates for the causal impact function.

Definition 18 (Causal Impact Function and Its Expectation). *Let $y^* \in \text{dom}(Y)$ denote the factual outcome value, and let $Y^s, Y^n \sim P_k^{s,n}$. For any measurable function $ci : \text{dom}(Y)^3 \rightarrow \mathbb{R}$, the (expected) causal impact of X_k is defined as*

$$\mathbb{E}[ci(Y^s, Y^n, y^*)] := \iint ci(y^s, y^n, y^*) P_k^{s,n}(dy^s, dy^n). \quad (4)$$

Informally, ci is a user-defined scoring rule that translates the joint sufficiency–necessity measure into a single scalar. The choice is not arbitrary: the Absolute Difference example below is an instance of L1 distance between outcomes, the same metric underlying Average Treatment Effect, Conditional Treatment Effect, and SHAP. The binary indicator score recovers this as a special case when outcomes are binary — indicators are L1 on $\{0, 1\}$. Practitioners with reasons to prefer L2 or another distance measure may substitute it freely; the framework places no restriction on ci beyond measurability.

For the OCB analysis we instantiate ci as the PNS binary score defined in the following example, with $y^* = \text{loan} = 0$. This is the choice that produces the values in Table 2.

Example 19 (PNS for Binary Outcomes). *Assume the outcome Y is binary with $\text{dom}(Y) = \{0, 1\}$, and let $y^* = 1$ denote the factual outcome. Let Y^s and Y^n denote the sufficiency and necessity outcome variables. Define the causal impact score*

$$ci(y^s, y^n, y^*) = \mathbb{I}\{y^n \neq y^*\} \mathbb{I}\{y^s = y^*\}.$$

Then the expected causal impact reduces to

$$\mathbb{E}[ci(Y^s, Y^n, 1)] = \mathbb{P}(Y^n = 0, Y^s = 1),$$

which conceptually coincides with Pearl’s probability of necessity and sufficiency (PNS). The alignment is exact when $\mathbf{S} = \{X_k\}$, $\mathbf{W} = \emptyset$, and $\text{dom}(X_k) = \{0, 1\}$ where, factually, $x_k^ = 1$.*

We now illustrate the full computation for Alice’s gender attribution using this ci . With $\mathbf{S} = \{\text{gender}, \text{credit}\}$, $\mathbf{W} = \{\text{check-failed}\}$, Γ_s and Γ_w uniform, Δ a point mass on the alternative binary value, and $y^* = \text{loan} = 0$, the exogenous noise U_{check} falls into three regions:

- u_1 (prob. 0.2): Alice is checked regardless of gender; $\text{check-failed} = 1$ factually.
- u_2 (prob. 0.7): Alice is not checked as female but would be as male; $\text{check-failed} = 0$ factually.
- u_3 (prob. 0.1): Alice is not checked regardless of gender.

In all regions, restoring Alice’s factual values always yields a denial, so $P_k^s(\{0\} \mid \mathbf{u}) = \frac{2}{3}$ everywhere (four suspect-containing (C, T) combinations each with $p^\Gamma(C, T) = \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$ — since $\mathbf{S} \cap \mathbf{W} = \emptyset$ here, rejection is vacuous and Γ is just the product of marginals — each contributing probability 1). The necessity measure varies.

In u_1 , the witness holds $\text{check-failed} = 1$. The four (C, T) contributions, each weighted $\frac{1}{6}$, are:

- $\mathbf{C} = \{\text{gender}\}, T = \emptyset$: male, $\text{check-failed} = 1$ propagates structurally $\Rightarrow P(\text{loan} = 1) = 0.05$.
- $\mathbf{C} = \{\text{gender}\}, T = \{\text{check-failed}\}$: male, witness holds failure $\Rightarrow P(\text{loan} = 1) = 0.05$.
- $\mathbf{C} = \{\text{gender}, \text{credit}\}, T = \emptyset$: male and good credit, check passes $\Rightarrow P(\text{loan} = 1) = 1.0$.

- $\mathbf{C} = \{gender, credit\}, T = \{check-failed\}$: male and good credit, but witness holds failure $\Rightarrow P(loan = 1) = 0.05$.

$$P_k^n(\{1\} \mid u_1) = \frac{1}{6}(0.05 + 0.05 + 1.0 + 0.05) \approx 0.192.$$

In u_2 , the witness holds $check-failed = 0$. The four contributions:

- $\mathbf{C} = \{gender\}, T = \emptyset$: male would be checked; $check-failed = 1$ propagates structurally $\Rightarrow P(loan = 1) = 0.05$.
- $\mathbf{C} = \{gender\}, T = \{check-failed\}$: male, witness holds $check-failed = 0$, check passes $\Rightarrow P(loan = 1) = 1.0$.
- $\mathbf{C} = \{gender, credit\}, T = \emptyset$: male and good credit, check passes $\Rightarrow P(loan = 1) = 1.0$.
- $\mathbf{C} = \{gender, credit\}, T = \{check-failed\}$: male and good credit, witness holds 0 $\Rightarrow P(loan = 1) = 1.0$.

$$P_k^n(\{1\} \mid u_2) = \frac{1}{6}(0.05 + 1.0 + 1.0 + 1.0) \approx 0.508.$$

In u_3 , male Alice is also not checked ($check = 0$ regardless of gender), so all necessity worlds yield denial: $P_k^n(\{1\} \mid u_3) = 0$.

Combining, with $ci(y^s, y^n, y^*) = \mathbb{I}\{y^n \neq y^*\} \mathbb{I}\{y^s = y^*\}$ and $y^* = loan = 0$:

$$\begin{aligned} \text{PCI}_{gender}(\text{Alice}) &= \mathbb{E}[ci(Y^s, Y^n, y^*)] = \mathbb{P}(Y^n = 1, Y^s = 0) \\ &= \sum_u p(u) P_k^s(\{0\} \mid u) P_k^n(\{1\} \mid u) \\ &= 0.2 \times \frac{2}{3} \times 0.192 + 0.7 \times \frac{2}{3} \times 0.508 + 0.1 \times 0 \\ &\approx 0.026 + 0.237 = 0.263. \end{aligned}$$

Table 2 gives the full results for Alice and Bob under these parameter choices, comparing PNS, PCI without witnesses, and PCI with the *check-failed* witness active.

Person	Feature	PNS	PCI (no witnesses)	PCI (with witnesses)
Alice	<i>gender</i>	0.045	0.210	0.263
Alice	<i>credit</i>	0.180	0.240	0.199
Bob	<i>gender</i>	0.000	0.038	0.019
Bob	<i>credit</i>	0.860	0.228	0.119

Table 2: Expected causal impact of *gender* and *credit* on $loan = 0$ for Alice (female, bad credit) and Bob (male, bad credit) in the stochastic OBCB model. PNS uses no suspect or witness structure; PCI without witnesses sets $\mathbf{W} = \emptyset$; PCI with witnesses sets $\mathbf{W} = \{check-failed\}$.

For Alice, PNS and PCI without witnesses both fail **D-A-rank** (p. 9): they rank *credit* above *gender* (0.180 vs. 0.045; 0.240 vs. 0.210). PCI with witnesses reverses this (0.263 vs. 0.199), satisfying **D-A-rank** as well as **D-A1** and **D-A2** (both values positive). Holding *check-failed* at its factual value blocks the credit-check bottleneck in noise realizations where Alice was checked, isolating gender’s structural role. The *check-failed* witness is the decisive ingredient.

For Bob, **D-B2** and **D-B-rank** hold under all three methods: *credit* is ranked above *gender* throughout. **D-B1** requires strictly positive attribution for *gender*: PNS returns exactly 0.000 and therefore fails, because $P(loan = 1 \mid gender = \text{female}, credit = \text{bad}) = 0$ makes the counterfactual necessity calculation zero, missing gender’s indirect role in opening the credit evaluation. PCI with witnesses returns 0.019, correctly capturing this indirect contribution. The cross-individual desideratum **D-comp** holds under both PCI (0.263 > 0.019) and PNS (0.045 > 0.000). A full verification of all seven desiderata is given at the end of this section.

The PNS and PCI (no witnesses) columns differ even though neither uses witnesses, because PCI still Γ -averages over candidate suspect sets. With $|\mathbf{S}| = 2$ PCI averages over $\{gender\}$, $\{credit\}$, and $\{gender, credit\}$ (each weighted $\frac{1}{3}$), whereas PNS evaluates a single counterfactual on one

variable at a time. Property (i) below establishes that the two agree exactly when $|\mathbf{S}| = 1$ (single suspect, no witnesses, binary outcome) — the OCB setup uses $|\mathbf{S}| = 2$ deliberately, to demonstrate joint-subset attribution.

The causal impact function ci is a user-defined component of PCI, and different choices encode different causal questions. In the benchmarking experiments of Section ??, we instantiate ci using only the necessity component, since Halpern’s actual causality has no sufficiency counterpart — necessity-only attribution allows a direct parallel comparison. The full framework incorporating both Y^s and Y^n is instantiated in the PNS binary example above and in the Absolute Difference example below; the OCB analysis of this section (Tables 2–3) provides a concrete validation, since the PNS binary ci jointly conditions on $Y^s = y^*$ and $Y^n \neq y^*$ and it is precisely this combination that produces the correct **D-A-rank** ordering. Further experiments using the full ci are presented in Section ??.

Add cross-reference to sufficiency experiments once written.

Example 20 (Absolute Difference Impact Score). *We can develop an analogous measure for continuous outcomes where we increase attribution with distance from the factual value y^* in the necessity world and decrease impact with distance to the factual value in the sufficiency world.*

$$ci(y^s, y^n, y^*) = |y^n - y^*| - |y^s - y^*|.$$

With the full machinery in place — structural model, variable selection, alternative value distribution, joint necessity-sufficiency measure, and causal impact function — PCI delivers the following properties.

- (i) **Generalisation of PN/PS/PNS.** When $\mathbf{S} = \{X_k\}$, $\mathbf{W} = \emptyset$, and $\text{dom}(X_k) = \{0, 1\}$, the PNS binary ci recovers Pearl’s probability of necessity and sufficiency exactly; PN and PS are recovered by the corresponding one-sided ci choices.
- (ii) **Connection to actual causality.** For suitable Γ and Δ , PCI recovers Halpern’s actual causality judgements (Section ??).
- (iii) **Necessity–sufficiency decomposition.** P_k^s and P_k^n are defined and interpretable independently; practitioners may choose a ci that uses either alone or both, depending on whether the causal question concerns necessity, sufficiency, or both.
- (iv) **Witness-mediated symmetry breaking.** Holding intermediate variables fixed via $\mathbf{W}$ disambiguates overdetermination and undercutting scenarios where necessity-only methods assign equal or zero attribution, as demonstrated above for Alice and Bob.
- (v) **Composability.** Γ , Δ , and ci are probability distributions and a measurable function respectively; they slot naturally into Bayesian pipelines, accept posteriors as inputs, and admit Monte Carlo estimation without modification to the core formalism.

Table 3 verifies the seven desiderata introduced at the start of this section (p. 9) against the values in Table 2. PCI with witnesses satisfies all seven; PCI without witnesses fails **D-A-rank**; PNS fails **D-A-rank** and **D-B1**. (The table uses the shorthand $|R(V \mid \text{person})|$ for $|R(V \rightsquigarrow \text{loan} \mid \text{person})|$.)

Desideratum	PNS	PCI (no wit.)	PCI (with wit.)
D-A1: $ R(\text{gender} \mid \text{Alice}) > 0$	✓ 0.045	✓ 0.210	✓ 0.263
D-A2: $ R(\text{credit} \mid \text{Alice}) > 0$	✓ 0.180	✓ 0.240	✓ 0.199
D-A-rank: $ R(\text{gender} \mid \text{Alice}) > R(\text{credit} \mid \text{Alice}) $	× 0.045 < 0.180	× 0.210 < 0.240	✓ 0.263 > 0.199
D-B1: $ R(\text{gender} \mid \text{Bob}) > 0$	× 0.000	✓ 0.038	✓ 0.019
D-B2: $ R(\text{credit} \mid \text{Bob}) > 0$	✓ 0.860	✓ 0.228	✓ 0.119
D-B-rank: $ R(\text{credit} \mid \text{Bob}) > R(\text{gender} \mid \text{Bob}) $	✓ 0.860 > 0.000	✓ 0.228 > 0.038	✓ 0.119 > 0.019
D-comp: $ R(\text{gender} \mid \text{Alice}) > R(\text{gender} \mid \text{Bob}) $	✓ 0.045 > 0.000	✓ 0.210 > 0.038	✓ 0.263 > 0.019

Table 3: Verification of the seven desiderata (p. 9) against the values in Table 2. PCI with witnesses satisfies all seven; PCI without witnesses fails **D-A-rank**; PNS fails **D-A-rank** and **D-B1**.

*PNS fails **D-A-rank** and **D-B1**. The D-A-rank failure reflects that without a check-failed witness, necessity-based attribution cannot distinguish Alice’s gender (which blocked the evaluation entirely)*

from her credit (which would only have mattered had she been evaluated). The D-B1 failure reflects that PNS assigns Bob’s gender exactly zero: because $P(\text{loan} = 1 \mid \text{female}, \text{bad}) = 0$, the counterfactual necessity calculation returns zero, missing gender’s indirect role in opening the credit evaluation. The check-failed witness resolves both.

To close this section, it is worth comparing PCI’s decomposition with the one already present in Pearl’s framework. Pearl’s framework defines three related but distinct binary scores: the probability of necessity $\text{PN} = P(Y_{x'} = y' \mid X = x^*, Y = y^*)$, the probability of sufficiency $\text{PS} = P(Y_{x^*} = y^* \mid X = x', Y = y')$, and their joint $\text{PNS} = P(Y_{x^*} = y^*, Y_{x'} = y')$. The first two are conditional probabilities; PNS is unconditional, and the three satisfy $\text{PNS} = \text{PN} \cdot P(Y_{x^*} = y^*) = \text{PS} \cdot P(Y_{x'} = y')$ rather than $\text{PNS} = \text{PN} \cdot \text{PS}$ (which would require $Y_{x^*} \perp Y_{x'}$, an assumption typically violated since the two potential outcomes share exogenous noise). PCI admits a related split: with $ci = \mathbb{I}\{y^n \neq y^*\} \mathbb{I}\{y^s = y^*\}$,

$$\mathbb{E}[ci(Y^s, Y^n, y^*)] = \int P_k^s(\{y^*\} \mid \mathbf{u}) P_k^n(\{1-y^*\} \mid \mathbf{u}) dP_{\mathbf{U}},$$

giving a *sufficiency marginal* and a *necessity marginal* that record how often, on average, restoring the factual values of a suspect set containing X_k reproduces the factual outcome, and how often substituting alternatives overturns it. The two marginals are independent given $\mathbf{u}$.

The components are not identical to Pearl’s: PN and PS condition on the factual outcome and consider a single counterfactual flip; PCI’s marginals integrate over $P_{\mathbf{U}}$, average over suspect sets weighted by Γ , and hold the witness configurations fixed when active. Table 4 shows both decompositions side by side for *gender* and *credit*.

Person, Variable	Pearl			PCI (with witnesses)		
	PN	PS	PNS	Necessity	Sufficiency	PCI
Alice (<i>gender</i>)	0.045	1.000	0.045	0.394	0.667	0.263
Alice (<i>credit</i>)	0.180	1.000	0.180	0.298	0.667	0.199
Bob (<i>gender</i>)	0.000	—	0.000	0.030	0.637	0.019
Bob (<i>credit</i>)	0.900	0.955	0.860	0.188	0.637	0.119

Table 4: Pearl’s PN/PS/PNS alongside PCI’s necessity and sufficiency marginals (with $\mathbf{W} = \{\text{check-failed}\}$ active). Bob’s PS for *gender* is undefined because the conditioning event (female, bad, loan = 1) has probability zero in the OBCB model. The PCI marginal counterparts are well-defined throughout, since they integrate over $P_{\mathbf{U}}$ rather than conditioning on the factual outcome.

For Alice’s gender, $\text{PN} = 0.045$ is the bare probability that a male with bad credit would have been approved; PCI’s necessity marginal is 0.394, nearly an order of magnitude higher, because the *check-failed* witness keeps the credit-check bottleneck active when the male counterfactual would otherwise have been checked and might still have failed — structural information that the unconditional PN cannot register. For Bob’s gender, PN is exactly 0 (a female with bad credit cannot be approved in this model) and PS is undefined; PCI’s marginals remain well-defined at 0.030 and 0.637, recovering the indirect role of gender in opening the evaluation.

The next section turns from comparing PCI with Pearl’s family of counterfactual scores to comparing it with the SHAP family — methods grounded in cooperative game theory rather than in structural counterfactual reasoning.

4 Examples and comparison to SHAP and Causal SHAP

Shapley values have become one of the most widely used tools for explaining the predictions of complex machine learning models. Grounded in cooperative game theory, they provide a principled decomposition of a model’s prediction into additive contributions from individual features. [Heskes et al. \[2020\]](#) propose Causal SHAP, which refines this by replacing the observational marginal used for features outside the active coalition with an interventional distribution: when a dropped feature is causally downstream of a coalition member, its distribution under the intervention differs from its

marginal, and Causal SHAP corrects for this using Pearl’s do-calculus. Formal definitions of both are given in Section 4.1.

Despite this repair, both methods share deeper limitations. First, both plain and Causal SHAP can assign non-zero responsibility to variables that play no causal role with respect to the target — attribution can run against the direction of the causal graph, a failure the interventional correction does not address. Second, both methods explain a model’s predicted output averaged over the population, not the individual factual outcome of a specific person: when the goal is *causal responsibility attribution*, this produces further failures that are not always made explicit in the literature.

We trace these failures through two examples, comparing plain SHAP, Causal SHAP, and PCI against the desiderata of Section 2 and some further desiderata related to the example we will employ. We first return to the OBCB model: it permits exact analytic computation and the desiderata are already known. In OBCB the feature set contains only exogenous roots, so plain and Causal SHAP coincide — the comparison is short. We then turn to a continuous mediation example where downstream variables are observable model inputs; there the two methods diverge, and we examine what Causal SHAP fixes and what it does not.

4.1 SHAP and Causal SHAP

Definition 21 (SHAP). *A cooperative game is a pair (N, v) , where $N = \{1, \dots, n\}$ is the set of players and $v : 2^N \rightarrow \mathbb{R}$ is the characteristic function mapping each subset (coalition) $S \subseteq N$ to the value that coalition can produce. The Shapley value [Shapley, 1953] is the unique attribution satisfying four axioms:*

1. **Efficiency:** $\sum_i \phi_i = v(N) - v(\emptyset)$
2. **Symmetry:** *If $v(S \cup \{i\}) = v(S \cup \{j\})$ for all S , then $\phi_i = \phi_j$*
3. **Dummy:** *If $v(S \cup \{i\}) = v(S)$ for all S , then $\phi_i = 0$*
4. **Linearity:** *For two games v_1, v_2 : $\phi_i(\alpha_1 v_1 + \alpha_2 v_2) = \alpha_1 \phi_i(v_1) + \alpha_2 \phi_i(v_2)$*

The unique solution is:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)] \quad (5)$$

The weight $\frac{|S|! (|N| - |S| - 1)!}{|N|!}$ equals the probability that, in a uniformly random ordering of all players, exactly the members of S precede i . Equivalently, ϕ_i is the average marginal contribution of i across all orderings. Lundberg and Lee [2017] apply Shapley values to explain model predictions by taking players to be input features, and the characteristic function to be the expected model output given a subset of features:

$$v(S) = \mathbb{E}_{X_{\bar{S}}} [f(X_S = x_S, X_{\bar{S}})] \quad (6)$$

where $\bar{S} = N \setminus S$ and missing features are marginalized over their marginal distribution.

Causal SHAP replaces the characteristic function with an interventional expectation [Heskes et al., 2020]:

Definition 22 (Causal SHAP).

$$v_{\text{causal}}(S) = \mathbb{E}[f(X) \mid \text{do}(X_S = x_S)] = \int P(X_{\bar{S}} \mid \text{do}(X_S = x_S)) f(X_{\bar{S}}, x_S) dX_{\bar{S}} \quad (7)$$

By the rules of do-calculus, intervening on X_S only affects the distribution of descendants of X_S . For non-descendants, $P(X_j \mid \text{do}(X_S = x_S)) = P(X_j)$. Writing $X_{\bar{S}, \text{nd}}$ for the features in $\bar{S}$ that are not descendants of any member of S , and $X_{\bar{S}, \text{d}}$ for those that are, equation (7) is equivalent to:

$$v_{\text{causal}}(S) = \mathbb{E}_{X_{\bar{S}, \text{nd}} \sim P(\cdot), X_{\bar{S}, \text{d}} \sim P(\cdot \mid \text{do}(X_S = x_S))} [f(X_{\bar{S}}, x_S)] \quad (8)$$

The Shapley formula (5) is otherwise unchanged. The method retains all four Shapley axioms.

4.2 OBCB: SHAP and Causal SHAP

To apply the SHAP and Causal SHAP definitions of Section 4.1 to OBCB we must specify both a feature set N and a function f on the joint feature space; the value functions and Shapley sums are derived from these. The natural choice for f is the conditional expectation of the outcome, $f(x_N) = \mathbb{E}[\text{loan} \mid X_N = x_N]$. But the OBCB SCM has five variables — *gender*, *credit*, *check*, *check-failed*, *loan-if-checked* — so the choice of N is non-trivial. Two options are natural:

- **Option A: two upstream roots.** $N = \{\text{gender}, \text{credit}\}$. Both features are exogenous; the three internal variables (*check*, *check-failed*, *loan-if-checked*) are integrated out inside f . This is the smallest sufficient feature set and matches the way the model is normally described: predict approval from gender and credit.
- **Option B: two roots plus the mediator.** $N' = \{\text{gender}, \text{credit}, \text{check-failed}\}$. The mediator becomes an observable model input, the same status it has in the signal-with-mediation example (§4.3). This is the comparison PCI’s witness mechanism naturally invites: PCI accesses *check-failed* as a witness, so allowing SHAP to access it as a feature is the fair foil.

We work through both. Option A is simpler and exposes the qualitative failures of plain SHAP. Option B then asks whether admitting the mediator rescues Causal SHAP.

Option A: two upstream roots. Here $f(g, c) = P(\text{loan} = 1 \mid g, c)$ takes values $f(\text{F}, \text{bad}) = 0$, $f(\text{F}, \text{good}) = 0.180$, $f(\text{M}, \text{bad}) = 0.045$, $f(\text{M}, \text{good}) = 0.900$, with the three internal variables integrated out. We apply SHAP to the rejection probability $g(x) = 1 - f(x)$ for Alice and Bob from Example 5, attributing responsibility for the denial outcome.¹² By the SHAP definition (and equivalently by Definition 22 once we note that no dropped feature is a descendant of any coalition member), the value function on the 2-feature game is

$$v(S) = \mathbb{E}[f(X) \mid X_S = x_S^*] = \sum_{x_{N \setminus S} \in \{0,1\}^{|N \setminus S|}} P(X_{N \setminus S} = x_{N \setminus S}) f(x_S^*, x_{N \setminus S}).$$

Under independent uniform marginals $P(\text{gender}) = P(\text{credit}) = \frac{1}{2}$ this specialises to

$$\begin{aligned} v(\emptyset) &= \frac{1}{4} \sum_{g,c \in \{0,1\}} f(g, c), & [\text{both features dropped: weight } P(g)P(c) = \tfrac{1}{2} \cdot \tfrac{1}{2}] \\ v(\{\text{gender}\}) &= \tfrac{1}{2}[f(g^*, 0) + f(g^*, 1)], & [\text{only credit dropped: weight } P(c) = \tfrac{1}{2}] \\ v(\{\text{credit}\}) &= \tfrac{1}{2}[f(0, c^*) + f(1, c^*)], & [\text{only gender dropped: weight } P(g) = \tfrac{1}{2}] \\ v(\{\text{gender}, \text{credit}\}) &= f(g^*, c^*). & [\text{nothing dropped: no average}] \end{aligned}$$

Plugging in the four values of f listed at the start of Option A yields Table 5; pushing those through the closed-form Shapley above gives the magnitudes in Table 6, which compares against PNS and PCI.

S	$v(S)$ for Alice	$v(S)$ for Bob
$\emptyset$	0.281	0.281
$\{\text{gender}\}$	0.090	0.473
$\{\text{credit}\}$	0.023	0.023
$\{\text{gender}, \text{credit}\}$	0.000	0.045

Table 5: Coalition values $v(S) = \mathbb{E}[f(X) \mid X_S = x_S^*]$ for the 2-feature game ($N = \{\text{gender}, \text{credit}\}$). $v(\emptyset) = \mathbb{E}[f(X)] = 0.281$ is the population baseline; $v(N)$ is the factual prediction $f(x^*)$. Plain SHAP and Causal SHAP coincide on this game (neither feature is downstream of the other), so a single column suffices per individual.

¹²Because $g = 1 - f$, the characteristic functions satisfy $v_g(S) = 1 - v_f(S)$ and $\phi_i^g = -\phi_i^f$; we compute ϕ^f and flip signs. With $|N| = 2$ and uniform marginals $P(\text{gender}) = P(\text{credit}) = \frac{1}{2}$, equation (5) reduces to $\phi_i = \frac{1}{2}[v(\{i\}) - v(\emptyset)] + \frac{1}{2}[v(N) - v(N \setminus \{i\})]$.

Person	Feature	PNS	SHAP	PCI (no witnesses)	PCI (with witnesses)
Alice	<i>gender</i>	0.045	0.107	0.210	0.263
Alice	<i>credit</i>	0.180	0.174	0.240	0.199
Bob	<i>gender</i>	0.000	−0.107	0.038	0.019
Bob	<i>credit</i>	0.860	0.343	0.228	0.119

Table 6: Responsibility attributions for Alice and Bob. SHAP values are signed (ϕ_i^g for $P(\text{loan} = 0 \mid x)$); desiderata compare magnitudes $|\cdot|$. PNS and PCI values from Table 2.

SHAP fails two desiderata. The primary failure is **D-A-rank**: Alice’s credit outranks her gender ($0.174 > 0.107$). This is an overdetermination effect: *credit*=bad correlates with the *check-failed* mechanism that blocked Alice’s credit evaluation, so SHAP’s observational marginalization inflates credit attribution above gender’s. PCI includes the factual *check-failed* value directly, isolating gender’s upstream causal role and recovering the correct ranking. PNS fails **D-A-rank** for the same reason; it also fails **D-B1** by returning exactly zero for Bob’s gender (see Table 3).

The second SHAP failure is **D-comp**: $|\phi_{\text{gender}}^g| = 0.107$ for both Alice and Bob. But gender played a strictly larger causal role in Alice’s rejection than in Bob’s. For Alice, *gender*=female was sufficient on its own to produce rejection — it blocked the credit evaluation entirely, so her credit was never a factor. For Bob, *gender*=male was a facilitating condition: it triggered the credit check, but *credit*=bad was the actual driver of the denial. SHAP cannot distinguish these because it measures how far each feature value deviates from the population prediction mean — female and male are equidistant deviations in opposite directions, so they receive equal magnitude regardless of their individual causal weight. This is an instance of a deeper limitation: SHAP explains predicted outputs, not individual factual outcomes, which we examine further in Section 4.3.

Causal SHAP coincides exactly with plain SHAP under Option A: both features are exogenous roots, so $P(\text{credit} \mid \text{do}(\text{gender} = g)) = P(\text{credit})$ and vice versa, no dropped feature is a descendant of any coalition member, and $v_{\text{causal}}(S) = v_{\text{plain}}(S)$ for every coalition. Option A therefore reduces to a single SHAP analysis, and the failures above belong to both methods.

Option B: two roots plus the mediator. Now we admit *check-failed* as a third model input and apply both methods to the 3-feature game $N' = \{\text{gender}, \text{credit}, \text{check-failed}\}$. This requires defining f on $\{0, 1\}^3$ rather than on $\{0, 1\}^2$. The natural observational extension is again the conditional expectation $\tilde{f}(g, c, cf) = \mathbb{E}[\text{loan} \mid \text{gender} = g, \text{credit} = c, \text{check-failed} = cf]$. This is well defined for three of the four (c, cf) combinations, but not the fourth: a failed check requires that a check happened *and* the credit was bad, so *check-failed* = 1 together with *credit* = good is impossible under the SCM and has no observational data to condition on. We leave $\tilde{f}$ undefined on this cell.

Both methods need a value function $v(S)$: the expected value of $\tilde{f}$ when the features in S are held at their factual values x_S^* and the remaining features are averaged out. The methods differ in how that average is taken. Plain SHAP averages over each dropped feature using its own observational marginal $P(X_i)$, with the dropped features treated as independent of one another. Causal SHAP averages over the joint distribution the SCM produces under the intervention $\text{do}(X_S = x_S^*)$, so the dropped features keep whatever joint dependencies the SCM imposes on them. Whether either average ever lands on the impossible cell ($c=1, cf=1$) — and so whether $v(S)$ is finite — depends on both the method and on which features are held fixed at which factual values. Table 7 works out all eight coalition values for both individuals.

Plain SHAP. Plain SHAP treats *credit* and *check-failed* as independent, so whenever both are dropped from the coalition the average runs over all four (c, cf) combinations, with the impossible ($c=1, cf=1$) combination receiving weight $P(c=1) \cdot P(cf=1) = 0.5 \cdot 0.275 \approx 0.14$ — the same nonzero weight as any other corner. For Alice, both c and cf are dropped at $S \in \{\emptyset, \{\text{gender}\}\}$, the first two NaN entries in Table 7. For Bob the same two coalitions fail, and two more break for an additional reason: at $S = \{\text{check-failed}\}$ and $S = \{\text{gender}, \text{check-failed}\}$ the value $cf^* = 1$ is

S	Alice $(g, c, cf)=(0, 0, 0)$		Bob $(g, c, cf)=(1, 0, 1)$	
	v_{plain}	v_{causal}	v_{plain}	v_{causal}
$\emptyset$	NaN	0.281	NaN	0.281
$\{g\}$	NaN	0.090	NaN	0.473
$\{c\}$	0.007	0.023	0.007	0.023
$\{cf\}$	0.270	0.270	NaN	NaN
$\{g, c\}$	0.000	0.000	0.014	0.045
$\{g, cf\}$	0.090	0.090	NaN	NaN
$\{c, cf\}$	0.000	0.000	0.025	0.025
N'	0.000	0.000	0.050	0.050

Table 7: Coalition values for the 3-feature game $N' = \{gender, credit, check-failed\}$. Bold **NaN** entries flag coalitions whose value function evaluates $\tilde{f}$ at the impossible cell $(c=1, cf=1)$ with positive weight. Plain SHAP and Causal SHAP coincide whenever neither *credit* nor *check-failed* is dropped, since the only structural coupling in the model is between these two; they diverge whenever one is in S and the other is not.

held fixed while c is sampled from its marginal, so the $c=1$ branch enters with weight 0.5 and again hits the impossible cell. Plain SHAP gives undefined Shapley values for both individuals.¹³

Causal SHAP for Alice. Causal SHAP samples cf following its structural equation $cf = check \cdot (1 - c)$, with $check \sim \text{Bern}(p_{\text{check}}[g])$, given whatever (g, c) values appear in the integrand. When $c = 1$ the equation forces $cf = 0$ regardless of $check$, so the $(c=1, cf=1)$ combination is never produced. Every Alice entry in the v_{causal} column of Table 7 is therefore finite. Plugging Alice’s column into the Shapley formula gives magnitudes $|\phi_{\text{gender}}| = 0.097$, $|\phi_{\text{credit}}| = 0.176$, $|\phi_{\text{check-failed}}| = 0.007$, so $|\phi_{\text{credit}}| > |\phi_{\text{gender}}|$ and **D-A-rank** fails: credit outranks gender, exactly the same pathology plain SHAP exhibited on the 2-feature game.

Causal SHAP for Bob. Two coalitions break for Bob: $S = \{check-failed\}$ and $S = \{gender, check-failed\}$. Both hold cf at Bob’s factual value $cf^* = 1$ and average *credit* over its marginal. To see why this is fatal, apply Definition 22 to the first:

$$v_{\text{causal}}(\{cf\}) = \mathbb{E}[\tilde{f}(X) \mid do(cf = 1)] = \frac{1}{4} \sum_{g \in \{0,1\}} \sum_{c \in \{0,1\}} \tilde{f}(g, c, 1).$$

The uniform $\frac{1}{4}$ weights come from *gender* and *credit* sitting upstream of cf in the causal graph: intervening on cf overrides the structural equation that would normally determine it from *credit* and *check*, but does not alter the marginal distributions of the upstream features. Two of the four summands evaluate $\tilde{f}$ at $(g, c=1, cf=1)$ — the impossible cell — with weight $\frac{1}{4}$ each. So $v_{\text{causal}}(\{cf\})$ is undefined, and so is $v_{\text{causal}}(\{g, cf\})$ for the same reason. These two coalitions appear in the Shapley sum for every feature, so all three of Bob’s Causal SHAP values are NaN (Bob’s columns in Table 7).

Why Bob breaks and Alice does not. The same two coalitions arise for Alice; the difference is the value of cf^* being held fixed. With Alice’s $cf^* = 0$, the $c=1$ summand in the formula above evaluates $\tilde{f}(g, 1, 0) = p_{\text{check}}[g] \cdot \text{loan_prob}[g, 0]$, which is defined because the SCM does allow $(c=1, cf=0)$: when *credit* is good, cf is forced to 0 deterministically, so the conditional expectation has a value. With Bob’s $cf^* = 1$, the $c=1$ summand evaluates $\tilde{f}(g, 1, 1)$, the impossible cell — “good credit and a failed check” — which the SCM rules out, since failing the credit check requires bad credit by the structural equation $cf = check \cdot (1 - c)$.

The mechanism producing this asymmetry is the intervention itself. When the SCM runs forward, *credit* and *check-failed* are linked by $cf = check \cdot (1 - c)$, which keeps $c=1$ and $cf=1$ from co-occurring. When Causal SHAP intervenes via $do(cf = cf^*)$, this link is overridden: cf is fixed externally and *credit* is sampled freely. The override always produces a $(c=1, cf^*)$ combination with positive weight; whether $\tilde{f}$ has a value for that combination depends on which cf^* value Causal SHAP is intervening on.

¹³The underlying issue is that plain SHAP does not see that *credit* and *check-failed* are linked by the structural equation $cf = check \cdot (1 - c)$, and so cheerfully samples a combination the SCM rules out.

Promoting the mediator to a feature therefore does not rescue Causal SHAP: it fails the desideratum on Alice and breaks down outright on Bob.

This example, however, has been a peculiar one. The OBCB SCM contains a structural equation tying *credit* and *check-failed* together, so Option B’s natural feature set ends up with one combination of input values — “good credit and a failed check” — that the model is not defined on. The impossible-cell phenomenon exposed real failure modes of plain SHAP and Causal SHAP, but it also obscured what the methods do on cleaner inputs. To see things more clearly we move to a simpler example, the signal-with-mediation chain $X \rightarrow M \rightarrow Y$, where the mediator is naturally an observable input, $\tilde{f}$ is defined everywhere, and no impossible-cell issue arises. Causal SHAP’s failures there are not artifacts of an undefined $\tilde{f}$ but properties of the method itself.

4.3 Signal with mediation

Example 23 (Signal with mediation). *We work with the following **generative model**: X sends a signal, M mediates it with some noise, and Y is the noisy final outcome:*

$$X \sim \mathcal{N}(0.5, 0.25) \tag{9}$$

$$M = X + \varepsilon_M, \quad \varepsilon_M \sim \mathcal{N}(0, 0.1) \tag{10}$$

$$Y = M + \varepsilon_Y, \quad \varepsilon_Y \sim \mathcal{N}(0, 0.1) \tag{11}$$

The causal graph is $X \rightarrow M \rightarrow Y$, a simple mediation chain. All noise terms are independent of each other and of X . Since everything is jointly Gaussian, all conditional expectations are linear functions of the conditioning variables — no approximations are required.

*Let $R(X \rightsquigarrow Y)$ denote the causal responsibility of X for Y . The **qualitative causal responsibility desiderata** for this example are:*

$$\mathbf{DXY} \quad R(X \rightsquigarrow Y) > 0$$

$$\mathbf{DMY} \quad R(M \rightsquigarrow Y) > 0$$

$$\mathbf{DXM} \quad R(X \rightsquigarrow M) > 0$$

$$\mathbf{DYX} \quad R(Y \rightsquigarrow X) = 0$$

$$\mathbf{DYM} \quad R(Y \rightsquigarrow M) = 0$$

$$\mathbf{DMX} \quad R(M \rightsquigarrow X) = 0$$

A somewhat more subtle desideratum is:

$$\mathbf{DMXY} \quad R(M \rightsquigarrow Y) > R(X \rightsquigarrow Y)$$

This captures the intuition that responsibility should be higher for variables that are closer to the outcome in the causal graph.

*The key **distributional facts** we will use are:*

$$\mathbb{E}[X] = \mathbb{E}[M] = \mathbb{E}[Y] = 0.5, \tag{12}$$

$$(\text{Var}(X), \text{Var}(M), \text{Var}(Y)) = (0.25, 0.35, 0.45), \tag{13}$$

$$(\text{Cov}(X, M), \text{Cov}(X, Y), \text{Cov}(M, Y)) = (0.25, 0.25, 0.35). \tag{14}$$

For each target variable, the optimal predictor under squared loss is the conditional expectation, computable exactly in this Gaussian model:¹⁴

$$f_Y(X, M) = \mathbb{E}[Y \mid X, M] = M \quad (\{X, M\} \Rightarrow Y) \quad (15)$$

$$f_M(X, Y) = \mathbb{E}[M \mid X, Y] = 0.5X + 0.5Y \quad (\{X, Y\} \Rightarrow M) \quad (16)$$

$$f_X(M, Y) = \mathbb{E}[X \mid M, Y] = 0.5 + \frac{5}{7}(M - 0.5) \quad (\{M, Y\} \Rightarrow X) \quad (17)$$

4.4 SHAP values for the signal with mediation example

First consider a situation in which $X = 1, M = 1, Y = 1$. For each of those outcomes we will check our causal responsibility attribution desiderata against SHAP values. First, let's compute the SHAP values for X and M with respect to the prediction of Y .

Target Y , Features $\{X, M\}$. First, compute the coalition values:

$$v(\emptyset) = \mathbb{E}[M] = 0.5 \quad (18)$$

$$v(\{X\}) = \mathbb{E}[M \mid X = 1] = 1 \quad (19)$$

$$v(\{M\}) = \mathbb{E}[M \mid M = 1] = 1 \quad (20)$$

$$v(\{X, M\}) = f_Y(1, 1) = 1 \quad (21)$$

Next, apply equation (5) with $|N| = 2$:

$$\phi_X = \frac{1}{2}(1 - 0.5) + \frac{1}{2}(1 - 1) = 0.25 \quad \phi_M = \frac{1}{2}(1 - 0.5) + \frac{1}{2}(1 - 1) = 0.25 \quad (22)$$

Target M , features $\{X, Y\}$. $v(\emptyset) = 0.5$; $v(\{X\}) = 1$; $v(\{Y\}) = 0.5 \mathbb{E}[X \mid Y=1] + 0.5 = 0.889$ (using $\mathbb{E}[X \mid Y=1] = 0.5 + \frac{0.25}{0.45}(0.5) = 0.778$); $v(\{X, Y\}) = 1$.

$$\phi_X = \frac{1}{2}(1 - 0.5) + \frac{1}{2}(1 - 0.889) = 0.306, \quad (23)$$

$$\phi_Y = \frac{1}{2}(0.889 - 0.5) + \frac{1}{2}(1 - 1) = 0.194. \quad (24)$$

Target X , features $\{M, Y\}$. $v(\emptyset) = 0.5$; $v(\{M\}) = 0.5 + \frac{5}{7}(0.5) = 0.857$; $v(\{Y\}) = \mathbb{E}[X \mid Y=1] = 0.778$ (as above); $v(\{M, Y\}) = \mathbb{E}[X \mid M=1] = 0.857$ (since $X \perp Y \mid M$).

$$\phi_M = \frac{1}{2}(0.857 - 0.5) + \frac{1}{2}(0.857 - 0.778) = 0.219, \quad (25)$$

$$\phi_Y = \frac{1}{2}(0.778 - 0.5) + \frac{1}{2}(0.857 - 0.857) = 0.139. \quad (26)$$

Summing up, the SHAP values for the three targets are as follows:

4.5 Causal SHAP for the signal example

We now apply Causal SHAP (Definition 22) to the signal example and compare against plain SHAP.

Target Y , features $\{X, M\}$. $v_{\text{causal}}(\emptyset) = 0.5$; $v_{\text{causal}}(\{X\}) = \mathbb{E}[M \mid \text{do}(X=1)] = 1$ (M is downstream of X : $P(M \mid \text{do}(X=1)) = \mathcal{N}(1, 0.1)$); $v_{\text{causal}}(\{M\}) = 1$ ($\text{do}(M=1)$ sets $f_Y = M = 1$ directly, $P(X \mid \text{do}(M=1)) = P(X)$ since X has no causal parents); $v_{\text{causal}}(\{X, M\}) = 1$. All values match plain SHAP, so $\phi_X = \phi_M = 0.25$.

¹⁴**Distributional facts.** From the structural equations: $\text{Var}(M) = \text{Var}(X) + \text{Var}(\varepsilon_M) = 0.25 + 0.1 = 0.35$; $\text{Var}(Y) = \text{Var}(M) + \text{Var}(\varepsilon_Y) = 0.35 + 0.1 = 0.45$. $\text{Cov}(X, M) = \text{Cov}(X, X + \varepsilon_M) = \text{Var}(X) = 0.25$; $\text{Cov}(X, Y) = \text{Cov}(X, M + \varepsilon_Y) = \text{Cov}(X, M) = 0.25$; $\text{Cov}(M, Y) = \text{Cov}(M, M + \varepsilon_Y) = \text{Var}(M) = 0.35$. **Conditional expectations.** $\mathbb{E}[Y \mid X, M] = M$ since $Y = M + \varepsilon_Y$ with $\varepsilon_Y \perp (X, M)$. For $\mathbb{E}[M \mid X, Y]$: the covariance matrix of (X, Y) is $\Sigma_{XY} = \begin{pmatrix} 0.25 & 0.25 \\ 0.25 & 0.45 \end{pmatrix}$ with $\det(\Sigma_{XY}) = 0.05$, so $\Sigma_{XY}^{-1} = 20 \begin{pmatrix} 0.45 & -0.25 \\ -0.25 & 0.25 \end{pmatrix}$; the regression coefficients are $[\text{Cov}(M, X), \text{Cov}(M, Y)] \Sigma_{XY}^{-1} = [0.25, 0.35] \cdot 20 \begin{pmatrix} 0.45 & -0.25 \\ -0.25 & 0.25 \end{pmatrix} = [0.5, 0.5]$, giving $\mathbb{E}[M \mid X, Y] = 0.5 + 0.5(X - 0.5) + 0.5(Y - 0.5) = 0.5X + 0.5Y$. For $\mathbb{E}[X \mid M, Y]$: by d -separation, $X \perp Y \mid M$ in the chain $X \rightarrow M \rightarrow Y$, so $\mathbb{E}[X \mid M, Y] = \mathbb{E}[X \mid M] = 0.5 + \frac{\text{Cov}(X, M)}{\text{Var}(M)}(M - 0.5) = 0.5 + \frac{0.25}{0.35}(M - 0.5) = 0.5 + \frac{5}{7}(M - 0.5)$.

Target	Method	ϕ_X	ϕ_M	ϕ_Y
Y	Plain SHAP	0.25	0.25	—
M	Plain SHAP	0.306	—	0.194
X	Plain SHAP	—	0.219	0.139

Table 8: Plain SHAP attributions for instance $X = 1, M = 1, Y = 1$. For target X , plain SHAP assigns $\phi_Y = 0.139 > 0$, even though Y does not cause X . This arises because $v(\{Y\}) = 0.778 > 0.5$: observing $Y = 1$ is statistically informative about $X = 1$ through the chain $X \rightarrow M \rightarrow Y$. SHAP cannot distinguish “ Y is informative about X ” from “ Y causes X ” because the value function is grounded in observational conditionals, which are symmetric under reversal of causal direction.

Target M , features $\{X, Y\}$. $v_{\text{causal}}(\emptyset) = 0.5$; $v_{\text{causal}}(\{X\}) = 1$ (Y is downstream of X , same as plain SHAP); $v_{\text{causal}}(\{Y\}) = \mathbb{E}_{X \sim P(X)}[0.5X + 0.5 \cdot 1] = 0.75$ (Y does not cause X , so $P(X \mid do(Y=1)) = P(X)$; differs from plain SHAP’s 0.889); $v_{\text{causal}}(\{X, Y\}) = 1$.

$$\phi_X = \frac{1}{2}(1 - 0.5) + \frac{1}{2}(1 - 0.75) = 0.375, \quad (27)$$

$$\phi_Y = \frac{1}{2}(0.75 - 0.5) + \frac{1}{2}(1 - 1) = 0.125. \quad (28)$$

Target X , features $\{M, Y\}$. $v_{\text{causal}}(\emptyset) = 0.5$; $v_{\text{causal}}(\{M\}) = 0.857$ (Y is downstream of M but f_X does not depend on Y since $X \perp Y \mid M$, so $\mathbb{E}_{Y \sim P(Y \mid do(M=1))}[f_X(1, Y)] = f_X(1, \cdot) = 0.857$; same as plain SHAP); $v_{\text{causal}}(\{Y\}) = \mathbb{E}_{M \sim P(M)}[f_X(M, 1)] = 0.5$ (M is not a descendant of Y , so $P(M \mid do(Y=1)) = P(M)$ and $\mathbb{E}[f_X(M, 1)] = 0.5 + \frac{5}{7}(\mathbb{E}[M] - 0.5) = 0.5$; differs from plain SHAP’s 0.778); $v_{\text{causal}}(\{M, Y\}) = 0.857$.

$$\phi_M = \frac{1}{2}(0.857 - 0.5) + \frac{1}{2}(0.857 - 0.5) = 0.357, \quad (29)$$

$$\phi_Y = \frac{1}{2}(0.5 - 0.5) + \frac{1}{2}(0.857 - 0.857) = 0. \quad (30)$$

4.6 PCI computation for the signal example

We work at the factual instance $X^* = M^* = Y^* = 1$, which gives abducted noise $\varepsilon_M^* = M^* - X^* = 0$ and $\varepsilon_Y^* = Y^* - M^* = 0$. The causal impact function is the **absolute difference score** (Example 20):

$$ci(y^s, y^n, y^*) = |y^n - y^*| - |y^s - y^*|.$$

Partial abduction: the factual noise is abducted in the sufficiency world; fresh independent noise is drawn in the necessity world. The alternative for the suspect is drawn from its marginal $P(A)$. For target B , $\mathbf{S}$ is the full set of two input features of B and $\mathbf{W}$ is the remaining input (neither suspect nor target); Γ_s is uniform over the three non-empty subsets of $\mathbf{S}$, and Γ_w is uniform over $2^{\mathbf{W}} = \{\emptyset, \{W\}\}$.

$R(X \rightsquigarrow Y)$. $\mathbf{S} = \{X, M\}$, $\mathbf{W} = \{M\}$, $w^* = m^* = 1$, $X' \sim P(X) = \mathcal{N}(0.5, 0.25)$. Rejection sampling from $\Gamma_s \times \Gamma_w$ accepts four of the six raw $(\mathbf{C}, \mathbf{T})$ pairs (Γ ’s normalizer $Z = 2/3$); each accepted pair carries weight $p^\Gamma(\mathbf{C}, \mathbf{T}) = \frac{1}{4}$, and the four sum to 1. Three of the four contain the suspect X and enter $P_X^{s,n}$; the fourth, $(\mathbf{C}=\{M\}, \mathbf{T}=\emptyset)$, carries the same weight $\frac{1}{4}$ but does not intervene on X , so it contributes 0 to the attribution. Per Definition 17, PCI is the Γ -integral of ci over the slice $\{X_k \in \mathbf{C}\}$ — not a renormalized expectation — so the three enumerated weights below sum to $\frac{3}{4}$, not 1.

Sufficiency (all three pairs): abducted $\varepsilon_Y^* = 0$ gives $Y^s = 1 = Y^*$, so $|Y^s - 1| = 0$ throughout. Sufficiency is maximally satisfied: the factual noise $\varepsilon_Y^* = 0$ means restoring X to its factual value exactly recovers the outcome.

Necessity by case:

$$\begin{aligned}
(C=\{X\}, T=\emptyset), \frac{1}{4}: & \text{ do}(X=X'); Y^n = X' + \varepsilon_M^n + \varepsilon_Y^n, \\
& Y^n - 1 \sim \mathcal{N}(-0.5, 0.45), \mathbb{E}[|Y^n - 1|] = 0.677. \\
(C=\{X\}, T=\{M\}), \frac{1}{4}: & \text{ do}(X=X', M=1); Y^n = 1 + \varepsilon_Y^n, \mathbb{E}[|Y^n - 1|] = \sqrt{0.2/\pi} \approx 0.252. \\
(C=\{X, M\}, T=\emptyset), \frac{1}{4}: & \text{ do}(X=X', M=M'); Y^n = M' + \varepsilon_Y^n, \\
& Y^n - 1 \sim \mathcal{N}(-0.5, 0.45), \mathbb{E}[|Y^n - 1|] = 0.677.
\end{aligned}$$

In the first and third cases M propagates freely; only the second case pins M as a witness and severs X 's indirect path. Combining:

$$R(X \rightsquigarrow Y \mid \mathbf{W}=\{M\}) = \frac{0.677 + 0.252 + 0.677}{4} \approx 0.402.$$

$R(Y \rightsquigarrow X)$. $\mathbf{S} = \{M, Y\}$, $\mathbf{W} = \{M\}$. X is exogenous with no parents, so any do-intervention leaves $X^n = X^* = 1$ (necessity) and $X^s = X^* = 1$ (sufficiency) in every (C, T) configuration. Hence $ci = 0$ for all pairs and

$$R(Y \rightsquigarrow X \mid \mathbf{W}=\{M\}) = 0.$$

4.7 Desiderata analysis for the signal example

Write ϕ_A^B for the attribution to feature A when predicting target B . Table 9 maps the desiderata from Section 4.3 to the values computed in Sections 4.4 and 4.5.

Desideratum	Condition	Plain SHAP	Causal SHAP	PCI ($\mathbf{W}=\emptyset$)	PCI ($\mathbf{W}=3\text{rd}$)
DXY	$R(X \rightsquigarrow Y) > 0$	0.250 ✓	0.250 ✓	0.452 ✓	0.402 ✓ [†]
DMY	$R(M \rightsquigarrow Y) > 0$	0.250 ✓	0.250 ✓	0.452 ✓	0.508 ✓
DXM	$R(X \rightsquigarrow M) > 0$	0.306 ✓	0.375 ✓	0.421 ✓	0.473 ✓
DYX	$R(Y \rightsquigarrow X) = 0$	0.139 ×	0.000 ✓	0.000 ✓	0.000 ✓
DYM	$R(Y \rightsquigarrow M) = 0$	0.194 ×	0.125 ×	0.000 ✓	0.000 ✓
DMX	$R(M \rightsquigarrow X) = 0$	0.219 ×	0.357 ×	0.000 ✓	0.000 ✓
DMXY	$R(M \rightsquigarrow Y) > R(X \rightsquigarrow Y)$	0.25=0.25 ×	0.25=0.25 ×	0.452=0.452 ×	0.508>0.402 ✓ [‡]

Table 9: Desiderata satisfied (✓) and violated (×) by plain SHAP, Causal SHAP, and PCI (alternative $\sim P(A)$, partial abduction) on the signal with mediation example ($X \rightarrow M \rightarrow Y$), instance $X = M = Y = 1$. PCI values are $\mathbb{E}[|y^n - B^*|]$; $\mathbf{S}$ = both input features of B , Γ_s/Γ_w uniform. **W=3rd**: the third variable (neither A nor B) is in $\mathbf{W}$ and sampled as an active witness when not in C . [†] 0.402 weights three suspect-containing (C, T) configurations at 1/4 each (the rejection-sampling normalizer $Z = 2/3$ also counts a fourth valid pair $(\{M\}, \emptyset)$ that does not contain X): one pins M as witness (contribution 0.252, X 's path severed) and two let M propagate freely (contribution 0.677 each). [‡] The two values use different witness sets ($\mathbf{W}=\{M\}$ for DXY, $\mathbf{W}=\{X\}$ for DMY), so the comparison is illustrative rather than a single-game ranking.

Positive results (DXY, DMY, DXM). Both methods correctly assign positive attribution to every direct or upstream cause of the target. Causal prediction models are built from observational conditionals, which do preserve correlation from cause to effect, so these desiderata pose no challenge to either method.

Causal SHAP fixes DYX, not DYM. For target X , the interventional value $v_{\text{causal}}(\{Y\}) = 0.5$ equals the baseline because M is not downstream of Y : $P(M \mid \text{do}(Y=1)) = P(M)$, so fixing $Y = 1$ provides no interventional leverage over f_X , and Causal SHAP correctly yields $\phi_Y^X = 0$. For target M , however, $\phi_Y^M = 0.125 > 0$ persists. The prediction model $f_M(X, Y) = 0.5X + 0.5Y$ contains Y as an input because Y is observationally predictive of M ; setting $Y = 1$ raises the model output even under the interventional regime. Causal SHAP replaces the conditioning distribution but cannot remove a feature that the prediction model itself uses.

DMX worsens under Causal SHAP. Plain SHAP assigns $\phi_M^X = 0.219$; Causal SHAP assigns 0.357. Fixing DYX reduces $v_{\text{causal}}(\{Y\})$ from 0.778 to 0.5, driving ϕ_Y^X to zero, but reallocates the

freed attribution to M . Since the structural noise that actually drives X is unobserved, all attribution concentrates on the only visible predictor, M , even though M does not cause X .

DMXY fails for both. Since $f_Y = M$, the singleton values $v(\{X\}) = \mathbb{E}[M \mid X=1] = 1$ and $v(\{M\}) = 1$ are identical, giving $\phi_M^Y = \phi_X^Y = 0.25$ for both methods. The Shapley formula cannot distinguish that X reaches Y only through M : when $X = 1$, the expected prediction is already 1 regardless of whether M is observed or not, so both features receive the same marginal contribution.

PCI: the role of the witness mechanism. Table 9 also reports PCI in two configurations: without witnesses ($\mathbf{W} = \emptyset$) and with the third variable as witness.

Without witnesses, PCI satisfies DYX, DYM, and DMX exactly — zero attribution to non-causes — because intervening on a variable that does not structurally affect the target leaves the target unchanged in both the necessity and sufficiency worlds, so $ci = 0$ identically. It also satisfies DXY, DMY, and DXM (all positive). However, **DMXY fails**: both $\text{PCI}(X \rightarrow Y \mid \mathbf{W}=\emptyset)$ and $\text{PCI}(M \rightarrow Y \mid \mathbf{W}=\emptyset)$ equal 0.452, because the path variance reaching Y is $\text{Var}(X) + \text{Var}(\varepsilon_M) + \text{Var}(\varepsilon_Y) = 0.45$ for X (through M) and $\text{Var}(M) + \text{Var}(\varepsilon_Y) = 0.45$ for M (directly) — an exact tie across every suspect coalition C , so the Γ_s -weighted average is also tied. Without a mechanism to block the indirect path $X \rightarrow M \rightarrow Y$, PCI cannot distinguish direct from indirect causation, mirroring the DMXY failure of both SHAP variants.

Admitting the third variable as a witness resolves this. $\text{PCI}(X \rightarrow Y \mid \mathbf{W}=\{M\}) \approx 0.402$ weights three suspect-containing (C, T) configurations at $1/4$ each: in two cases M propagates freely (contributing 0.677 each) and in one case M is held fixed as witness, severing X 's indirect path (contributing 0.252). $\text{PCI}(M \rightarrow Y \mid \mathbf{W}=\{X\}) \approx 0.508$ weights three cases in which M takes an alternative value; since X does not appear in Y 's structural equation, pinning X as a witness has no effect on Y^n , and all three cases contribute 0.677. The gap $0.508 > 0.402$ satisfies DMXY: the witness mechanism selectively reduces X 's attribution (via the M -witness case) while leaving M 's unchanged. The comparison relies on different witness sets for the two cells, so it is illustrative rather than a single-game ranking.

Stepping back from the individual desiderata, the failures above are not artifacts of a particular feature set or parameter choice: each traces to a structural limitation of the SHAP/Causal SHAP framework. Causal SHAP strictly improves on plain SHAP for one desideratum (DYX), by replacing observational conditioning with interventional distributions. The remaining failures share two sources. Prediction models include non-causal features that the conditioning distribution cannot exclude (DYM, DMX). Shapley averaging cannot distinguish direct from indirect causal paths (DMXY). PCI addresses both through the structural intervention framework and the witness mechanism.

Read together with the OCB analysis of Section 4.2 — where plain SHAP fails **D-A-rank** on the 2-feature game, and Causal SHAP either inherits that failure (Alice) or becomes undefined outright (Bob) once the mediator is admitted as a feature — the picture across both examples is the same: neither SHAP variant reliably satisfies the desiderata that motivated PCI in Section 2. The witness mechanism, the suspect-set distribution Γ_s , and the use of the realized outcome rather than the prediction baseline are the three ingredients of PCI that the SHAP framework systematically lacks; the failures catalogued above are the shape of that absence.

5 Synthetic evaluation with overdetermination and undercutting

In this evaluation, we build off the existing counterexamples to the but-for clause as an explanation tool. Recall that the initial intuition one might have is the following:

$X = x$ is responsible for $Y = y$ just in case Y would have been different had we intervened on X to be different, i.e. if $Y_{x'} = y'$ for some $x' \neq x, y' \neq y$.

A typical counterexample to this is cast in terms of categorical variables Halpern [2016]:

Sally and Billy pick up stones and throw them at a bottle. Sally's stone gets there first, shattering the bottle. Both throws are perfectly accurate, so Billy's stone would have shattered the bottle had it not been preempted by Sally's throw. The intuition is that Sally's throw is responsible for the shattering, but not so Billy's.

However, it is false that had Sally not thrown the stone, the bottle would not have shattered, so the simple counterfactual condition fails.

What we want to investigate is the performance of responsibility attribution, using the same implementation as the one used in Appendix ??, following the discussion in Section ??, in the context of a synthetic generalization of this problem. Suppose we use continuous variables in a model with a similar overdetermination and undercutting structure. Is the tool capable of meeting our expectations when it comes to responsibility attribution?

The answer is positive. We first introduce the model, introduce intuitive qualitative desiderata, and investigate their satisfaction for 500 local explanations, for events generated using the same generative process, with responsibility scores for all features estimated using one sample of size 5000.

The way we generalize is as follows. The input and output variables are continuous. The A variables are like Sally’s throw: they have an unblocked impact on the outcome. The C variables are like Billy’s throw: if their linear combination is high enough, they are active and can have an impact on the outcome. Otherwise, their impact is blocked. B and D are like ‘Sally hits’ and ‘Bill hits’, except continuous, and E is some final outcome, except now we don’t use a disjunction, but rather addition.

For $i = 0, 1, 2$:

$$A_i \sim \mathcal{N}(0, 1),$$

$$C_i \sim \mathcal{N}(0, 1),$$

and

$$B = 0 \cdot A_0 + 5 \cdot A_1 + 10 \cdot A_2,$$

$$cs_dot = 0 \cdot C_0 + 5 \cdot C_1 + 10 \cdot C_2,$$

$$cs_win = \mathbf{1}\{cs_dot > B\},$$

$$D = cs_dot \cdot cs_win,$$

$$E = B + D.$$

See Figure 1 for a visualization of the model structure.

Say we denote the mean causal impact of a variable V by $ci(V)$. We do not expect to see zero estimates for irrelevant properties, as the score also results from sampling other potential causes and witnesses, so there will always be some distance between the outcomes in both sufficiency and necessity worlds. But we do expect the following qualitative/comparative statements to hold:

Most of the time, $ci(A0) < ci(A1) < ci(A2)$, with $ci(A1)$ and $ci(A2)$ closer to each other and potentially the $ci(A1) > ci(A2)$ for cases where $A1 \gg A2$.

Similarly, for the C variables, when their impact is not blocked (i.e. if ‘ $cs_win = 1$ ’). Otherwise, we expect $ci(C1) \approx ci(C2) \approx ci(C3)$. Given the difference in means between $ci(A0)$ and $ci(A2)$ close to 6, let’s explicate this proximity as the absolute difference between sampled values being < 2 . Mean estimated scores for all the features involved and the frequency of the binary claims satisfaction are as expected, and can be inspected in Figures 2 and 3

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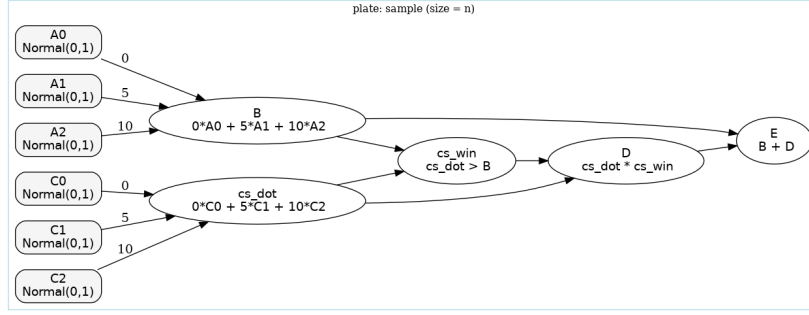


Figure 1: Synthetic model involving overdetermination and undercutting used in the evaluation loop.

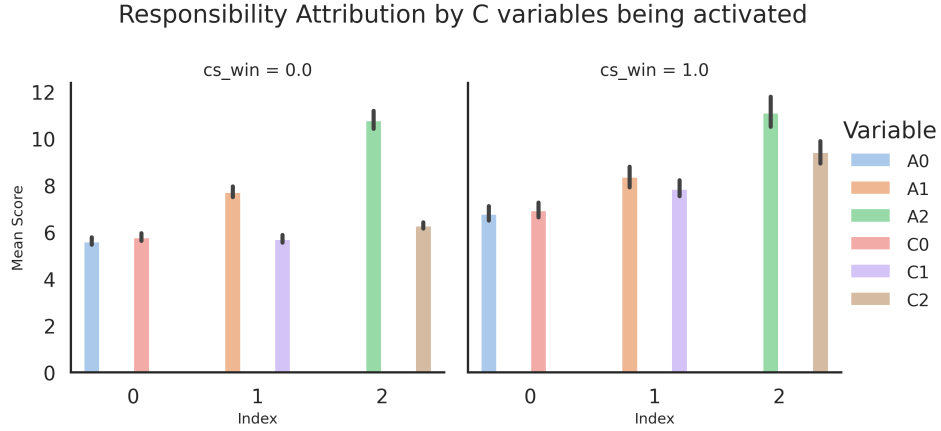


Figure 2: Mean attribution scores to all upstream features in the synthetic model, 500 individual explanations, all based on a single search sample of size 5000. When ‘cs_win=0’, the C variables are not activated, and so we expect total scores for all of them to be similar (red, violet, dark brown on the left-hand side). We still expect the scores for $A0$, $A1$, and $A2$ generally to increase independently of what happens with the C variables, which we see on both sides (blue, light brown, and green on both sides). We also expect the scores for C variables to increase if they are activated (red, violet, dark brown on the right-hand side). These expectations are satisfied.

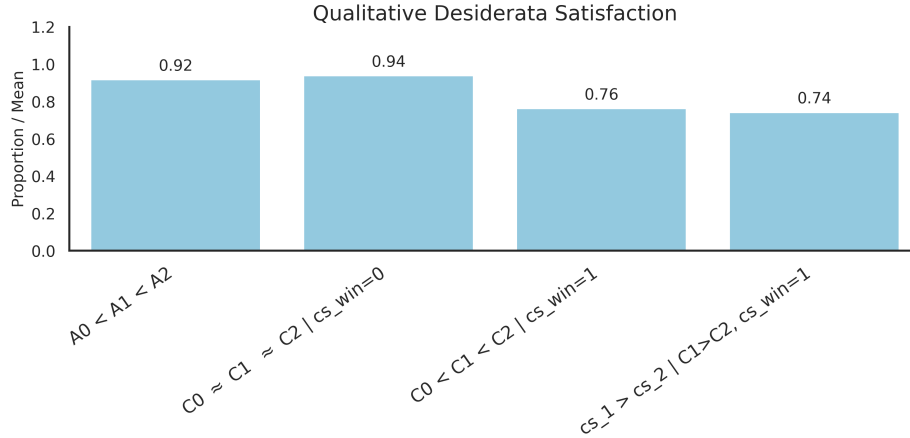


Figure 3: Qualitative claims satisfaction frequency for causal attribution in the synthetic model, 500 local explanations all based on a single search sample of size 5000. Most of the time, the ordering of the scores for A variables agrees with the ordering of the coefficients. Most of the time, close proximity of scores for inactive C variables is quite high; most often, the ordering for active C variables agrees with the ordering of the coefficients. We do not, however, want an explanation to track coefficients only, which, for instance, is illustrated by the last statement: if scores are reversed, this is mostly due to the disparate direction of the difference between the variable values.

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